

Permanent Labor Income Inequality and Stabilization Policies

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Abstract

We study how rising permanent labor income (PLI) inequality changes the transmission of monetary and fiscal policy. Standard heterogeneous-agent models imply that hand-to-mouth status is unrelated to permanent income, and that the PLI distribution is therefore neutral. In the PSID, instead, the hand-to-mouth share falls from 64% to 22% across PLI quintiles, a gradient that has steepened since 1999. A model with a non-homothetic taste for wealth reproduces these facts and breaks the neutrality. The incidence of fiscal policy becomes a first-order determinant of the output response: the same deficit stimulates or contracts output depending on who receives and who repays it. For monetary policy, the effect of a rate cut depends on how the government rebates its interest savings. The rise in PLI inequality since 1970 also raised the aggregate MPC from 0.30 to 0.36, strengthened tax-financed fiscal packages while weakening transfer-financed ones, and made monetary policy slightly stronger, more regressive, and more dependent on the fiscal rule.

Keywords: Inequality, permanent income, monetary policy, fiscal policy, interest rates, heterogeneous agent model.

JEL codes: D63, E21, E24, E52, E62

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Introduction

Labor-income inequality in the United States has risen sharply since 1980, with the share of pre-tax labor income earned by the top one percent rising from 8% to 14% (Piketty et al. 2018, Saez & Zucman 2020). Most of this rise comes from the permanent component of earnings: gaps in lifetime earnings across workers have widened, while the volatility of individual careers has not increased (DeBacker et al. 2011, Bloom et al. 2017, Braxton et al. 2021, Guvenen et al. 2022). We ask: does this change in the distribution of permanent labor income (PLI) affect the transmission of monetary and fiscal policy?

In standard heterogeneous-agent models, the answer is no. Consumption is linear in permanent income (Straub 2019), and this linearity has a strong aggregate implication¹: in a Heterogeneous-Agent New-Keynesian (HANK) model with homothetic preferences, every sequence-space Jacobian of the household block, and with it the output response to any monetary or fiscal shock, is invariant to the distribution of permanent labor income. In these models, four decades of rising permanent inequality have no effect on stabilization policy.

However, the data reject a key implication of these models: that hand-to-mouth status is unrelated to permanent labor income. In the PSID, the share of hand-to-mouth households falls from 64% in the bottom quintile of permanent labor income to 22% in the top one, and transitions in and out of hand-to-mouth status show that this is a persistent characteristic rather than a temporary spell. This gradient has steepened as inequality rose: between 1999 and 2019, the net-worth hand-to-mouth gap between low- and high-permanent-income households widened from 13 to 24 percentage points, and the slope of the relationship nearly doubled. These facts extend the saving-rate evidence of Dynan et al. (2004), Straub (2019): households with higher permanent in-

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¹We state this result as Proposition 4 in Appendix A.

come save more of it, and households with low permanent income increasingly live hand to mouth.

We develop a HANK model in which these facts arise from one departure from the benchmark: households have a non-homothetic taste for wealth, so that wealth is a luxury good (Carroll 1998, Kumhof et al. 2015, Michau 2018, Straub 2019, Mian et al. 2021). In this model, high-permanent-income households save a large share of their income, because they value wealth for its own sake and not only as a buffer against shocks. Their savings push down the equilibrium interest rate. At the bottom of the distribution, the taste for wealth is negligible, since consumption levels are low. Facing a low interest rate, these households dissave toward the borrowing constraint. The share of hand-to-mouth households then falls steeply with permanent income, as in the data.

Calibrated to the U.S. in 2019, the model matches the Pareto tail of wealth, the level of aggregate wealth, and the aggregate hand-to-mouth share, a combination that is difficult for one-asset HANK models (Kaplan et al. 2018, Auclert et al. 2024), and it reproduces the hand-to-mouth gradient of the PSID. We compare it to a heterogeneous-discount-factor economy (het-beta) that matches the same aggregates with a flat gradient. Since het-beta is homothetic, every difference between the two models comes from the correlation between MPCs and permanent labor income.

The model has two sets of implications for stabilization policy. First, holding the PLI distribution fixed, the wedge between the MPC out of labor income and the MPC out of a transfer is larger in our baseline model than under the homothetic benchmark model: labor income accrues disproportionately to households far from the borrowing constraint, so a dollar of labor income generates 20% less spending in present value than a transfer dollar, against 2% in het-beta. The incidence of fiscal policy then becomes a first-order determinant of the cumulative multiplier. Holding the deficit path fixed, switching the instrument that repays it moves each cumulative multiplier by more than 0.5 in the baseline, against less than 0.25 in het-beta, and can flip its sign: a tax cut repaid by future transfer cuts has a cumulative multiplier of -0.61 , as it hands dollars to households who save them and takes them from households who spend them.

The same wedge carries to monetary policy. The direct effect of a rate change on consumption is nearly identical in the two models; the difference lies in the fiscal side. A rate cut lowers the interest bill on public debt, and the output response depends on how the government rebates this windfall: switching the rebate from taxes to transfers changes the output response by 98% in the baseline, against 14% in het-beta. The two economies share the same hand-to-mouth share, the same wealth concentration, and nearly the same aggregate MPC; they transmit policy differently because the identity of the hand-to-mouth households differs.

Second, a rise in PLI inequality now affects the transmission of both instruments. We re-solve the baseline model under the 1970 PLI distribution and compare the two steady states. Through the lens of our model, the observed increase in labor income inequality raised the hand-to-mouth share from 0.31 to 0.39, the aggregate annual MPC from 0.30 to 0.36, and lowered the aggregate elasticity of intertemporal substitution from 0.81 to 0.75. Between two thirds and four fifths of each change comes from the general-equilibrium fall in the real rate produced by the extra saving of high-permanent-income households. Fiscal policy becomes stronger when repaid through taxes and weaker when repaid through transfer cuts: the transfer payout repaid by taxes rises from 0.82 to 0.93, while the tax cut repaid by transfer cuts falls from -0.49 to -0.61 . The timing of the adjustment matters more as well: postponing the repayment of a deficit raises the cumulative multiplier in both economies, but by more under the 2019 distribution, since with more households near the constraint, future repayments are discounted more heavily; the gap reaches about 0.4 for tax-financed and 0.5 for transfer-financed deficits as repayment moves far into the future. The output response to a monetary shock grows by about 4%, but its composition changes: the direct rate channel shrinks, the income channels expand by a fifth, the valuation gains of the top 1% grow by about a third, and the response becomes about 20% more sensitive to the fiscal rule. In het-beta, the same experiment leaves all of these numbers unchanged.

Related Literature. Our paper makes three contributions. First, we document that hand-to-mouth status is concentrated among households with low permanent labor income, and that this concentration has increased since 1999. Second, we extend the intertemporal Keynesian cross to make the incidence of fiscal and monetary policy

explicit, and we generalize the aggregate neutrality result of Straub (2019): under homothetic preferences, every sequence-space Jacobian, and hence the transmission of any policy shock, is invariant to the PLI distribution. Third, we quantify the consequences: under the 2019 distribution, the incidence of fiscal and monetary policy is a first-order determinant of their effects, and the rise in PLI inequality since 1970 has changed the transmission of both instruments.

Monetary policy transmission with heterogeneous households. Micro data show that a large share of households hold little liquid wealth (Kaplan et al. 2014). With such households, most of monetary transmission runs through indirect general-equilibrium effects (Kaplan et al. 2018), redistribution becomes a transmission channel (Auclert 2019, Coibion et al. 2017), the fiscal response to a monetary shock matters for its effect (Alves et al. 2020), and the indirect effects arrive slowly (Holm et al. 2021). This literature characterizes transmission along the dimensions of wealth and liquidity; we add the dimension of permanent labor income. Closest in mechanism, Patterson (2023) shows that high-MPC workers are more exposed to business-cycle earnings fluctuations, and that this covariance amplifies recessions; we study the same covariance along the permanent component of earnings, where it is negative, and where its size depends on the PLI distribution. On the empirical side, the closest paper is Aguiar et al. (2025), who characterize who the hand-to-mouth households are in the PSID. We add that hand-to-mouth status lines up with permanent labor income, and that this relationship has tightened as earnings inequality rose.

The intertemporal Keynesian cross. Auclert et al. (2024) show that the output response to a fiscal shock is priced by the intertemporal MPCs of the households who receive and repay it, and Auclert et al. (2025) survey the framework’s applications to fiscal and monetary policy. We generalize the cross in two directions. First, the benefit and financing incidence of every instrument enters explicitly, so that the same equation prices deficit-financed packages, balanced-budget redistributions, and any combination of payout and repayment instruments. Second, the government budget carries the cross over to monetary policy: the fiscal windfall created by a rate cut is itself a package, and its incidence is set by the fiscal rule.

Non-homothetic taste for wealth. Our saving motive comes from Straub (2019), who doc-

uments that saving rates rise with permanent income, proposes a non-homothetic taste for wealth to match this, and traces its consequences for wealth inequality and the long-run interest rate. A broader literature uses similar preferences: Carroll (1998) for the saving rates of the rich, Kumhof et al. (2015) for the build-up of household debt before the Great Recession, Michau (2018) and Mian et al. (2021) for the over-accumulation of savings, and Gaillard et al. (2023) for the relative tails of the income and consumption distributions.² This literature uses non-homotheticity to explain long-run trends in saving and wealth; we bring it to stabilization policy. Sharing our question, Melcangi & Sterk (2020) find that rising stock-market participation has strengthened U.S. monetary policy since 1980; we study the change in transmission produced by the labor-income distribution itself.

Outline. Section 1 documents the two PSID facts. Section 2 presents the model, and Section 3 its calibration. Section 4 derives the incidence-explicit IKC and compares the baseline model with het-beta under the 2019 distribution. Section 5 measures the effect of rising PLI inequality on fiscal and monetary transmission.

1 Permanent Labor Income and Hand-to-Mouth Status

This section documents two empirical facts using the Panel Study of Income Dynamics (PSID, 1999–2019). First, hand-to-mouth (HtM) status is strongly concentrated among households with low permanent labor income. Second, this concentration has increased over the sample period.

Data and definitions. We use the PSID’s biennial waves from 1999 to 2019, covering 33,498 household-year observations from 5,565 households. Each wave records income for the preceding calendar year, so statistics dated by income year run from 1998 to 2018.³ We follow Aguiar et al. (2025) and classify a household as *hand-to-mouth in net worth* (HtM_{NW}) if its net worth is below two months of household earnings. We clas-

²De Nardi et al. (2016) survey the related evidence on bequest motives.

³Sample selection follows Aguiar et al. (2025): heads aged 25–64, real labor earnings and disposable income above \$2,000, and at least three waves observed per household (required to measure permanent income). The household earnings measure, taken from the Aguiar et al. (2025) pipeline, sums head and spouse labor income, government transfers, and employer payroll contributions; the same measure is the denominator of both hand-to-mouth thresholds below. All dollar amounts are deflated with the PCE price index (2012 base). Statistics are weighted by PSID family weights.

sify a household as a *wealthy hand-to-mouth* (HtM_{LIQ}) if its liquid assets fall below one week of income, but its net worth is above the hand-to-mouth in net worth threshold, following the implementation in Aguiar et al. (2025) of the wealthy hand-to-mouth concept of Kaplan et al. (2014). Appendix C reports summary statistics by permanent income quintile.

Our baseline measure of permanent labor income (PLI) follows Straub (2019). We regress log household labor earnings on an age cubic, family size, a dual-earner indicator, and year fixed effects, then average the residuals within each household across all available waves⁴. Permanent differences account for a large share of residual earnings dispersion: a minimum-distance decomposition of the residual process (Appendix D) attributes about half of the cross-sectional variance of residual log earnings ($\hat{\sigma}_\alpha^2 = 0.28$ out of 0.55) to a fixed household component, with the remainder split between a persistent AR(1) component (annual persistence 0.91) and transitory variation and measurement error. Appendix B shows that the facts below are robust to different measures of permanent labor income.

Fact 1: HtM status declines steeply with permanent income. Panel (a) of Figure 1 presents HtM shares by PLI quintile. The HtM_{NW} share falls monotonically from 38% in Q1 (lowest permanent income) to 12% in Q5 (highest). The pattern is similar for HtM_{LIQ} , declining from 26% to 10%. Summing the two mutually exclusive categories, the total hand-to-mouth share falls from 64% to 22%. A household in the bottom quintile of permanent income is more than three times as likely to be hand-to-mouth in net worth as one in the top quintile.

Panel (a) shows a stock: the share of households that are hand-to-mouth at a point in time. A stock alone cannot tell whether the same households remain hand-to-mouth over time. Panel (b) therefore shows the flows: four-year transition probabilities into and out of hand-to-mouth status by PLI quintile, summing over the two definitions of hand-to-mouth, as they are mutually exclusive. Among households that are hand-to-mouth in wave t , the probability of remaining hand-to-mouth four years later (wave $t + 2$) falls from 78% in Q1 to 50% in Q5. Among households that are *not* hand-to-

⁴This measure includes richer first-stage controls than the PLI measure of Aguiar et al. (2025) (which conditions only on an age cubic and year), though the two are highly correlated (correlation 0.89).

mouth in wave t , the probability of becoming hand-to-mouth falls from 39% in Q1 to 11% in Q5. The flows show the same gradient as the stock: low-PLI households enter hand-to-mouth status more often and exit it less often, so the cross-sectional concentration reflects a persistent characteristic rather than temporary spells.

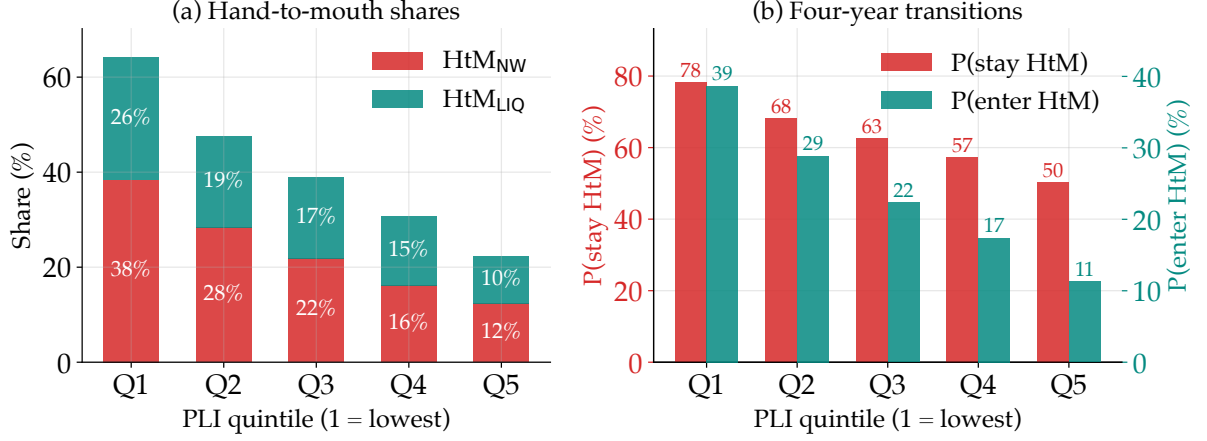


Figure 1: Hand-to-mouth status by PLI quintile: stocks and flows (PSID, 1999–2019)

Notes: Panel (a): HtM_{NW} = net-worth hand-to-mouth: net worth below two months of earnings (Aguiar et al. 2025). HtM_{LIQ} = wealthy hand-to-mouth: net worth above the threshold but liquid assets below one week of income (Kaplan et al. 2014). Shares are computed among households with non-missing hand-to-mouth status. Panel (b): “P(stay HtM)” = probability of being hand-to-mouth (HtM_{NW} or HtM_{LIQ}) in wave $t + 2$ (four years later) conditional on being hand-to-mouth in wave t (left axis); “P(enter HtM)” = probability of becoming hand-to-mouth conditional on not being hand-to-mouth in wave t (right axis). PLI quintiles computed from the averaged residuals of Straub (2019). Weighted by PSID family weights.

Fact 2: The HtM–PLI gradient has steepened since 1999. Panel (a) of Figure 2 plots HtM_{NW} shares over time for low-PLI (Q1–Q2) and high-PLI (Q4–Q5) households, together with the aggregate share. The aggregate HtM_{NW} share rose from 18% in 1999 to 28% in 2012 before declining modestly. This aggregate increase, however, masks a divergence across the PLI distribution. The HtM_{NW} share among low-PLI households rose from 25% to 36%, while the share among high-PLI households ended the sample at 12%, where it started, after a temporary rise around the Great Recession. The gap between the two groups thus widened from 13 to 24 percentage points over two decades.

Panel (b) shows the same steepening without binning households into quintiles. For each wave t we estimate the weighted linear probability model

$$\text{HtM}_{i,t} = a_t + b_t \widehat{\text{PLI}}_i + u_{i,t}, \quad (1)$$

where $\text{HtM}_{i,t}$ is an indicator for HtM_{NW} status and $\widehat{\text{PLI}}_i$ is the household’s averaged

log-earnings residual, which is fixed over time. The slope b_t measures the change in the probability of being hand-to-mouth per log point of permanent labor income; because $\widehat{\text{PLI}}_i$ does not vary over time, changes in b_t reflect changes in who is hand-to-mouth, not in the permanent income measure itself. The gradient nearly doubles over the sample, from -0.10 to -0.19 . The gradient steepens both before 2008 and after 2010, so the divergence in Panel (a) is not a one-off legacy of the 2008–2009 recession and housing bust. The steepening also lines up with inequality itself: interacting PLI with the wave-level dispersion of residual earnings yields a negative and strongly significant coefficient (Table 5 in Appendix C).

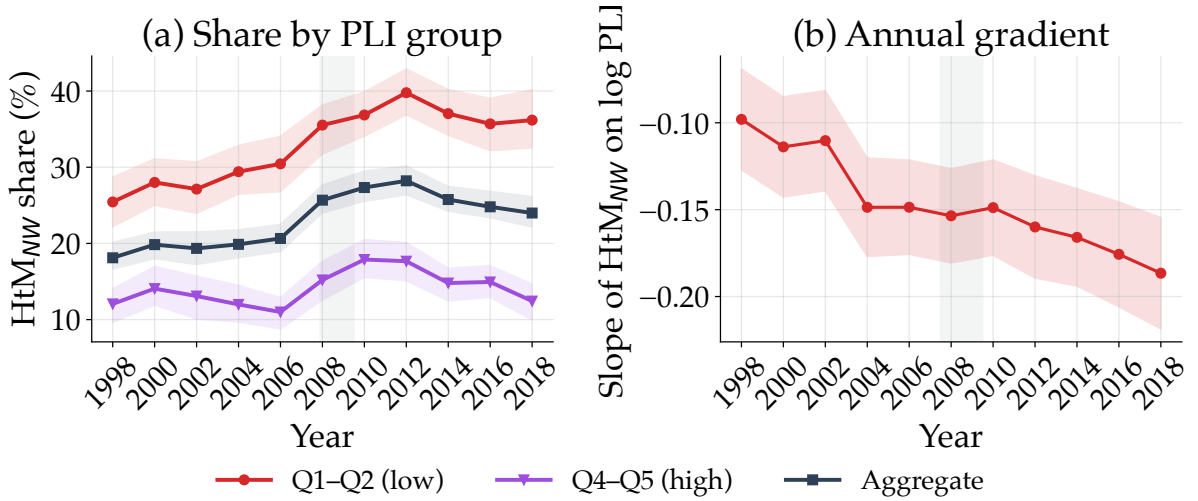


Figure 2: The steepening of the HtM–PLI gradient (PSID, 1999–2019)

Notes: Panel (a): weighted HtM_{NW} shares by PLI group; shaded bands are 95% bootstrap confidence intervals. Panel (b): slope coefficients of weighted linear probability regressions of HtM_{NW} status on log PLI, estimated separately by wave; the shaded band is the 95% confidence interval (robust standard errors). PLI groups (Panel a) and the continuous PLI measure (Panel b) are based on the averaged residuals of Straub (2019). The vertical grey bar marks the Great Financial Crisis (2008–2009).

2 A HANK Model with Permanent Labor Income Inequality

The previous section established that there exists a strong PLI–HtM gradient in the data. We now introduce a model that can reproduce this feature. The model is a canonical one-asset HANK with rigid wages, augmented with two ingredients: *ex ante* heterogeneity in permanent labor income and a non-homothetic taste for wealth. Section 3 discusses how these ingredients match the gradient.

The economy is populated by a continuum of households, a continuum of intermediate-

good firms, a representative final good producer, a labor union, a mutual fund, a government, and a central bank. Time is discrete and denoted t . There is no aggregate risk: monetary and fiscal shocks are MIT shocks,⁵ and a change in the distribution of permanent labor income is a comparison across steady states.

2.1 Households

Households save in a mutual fund, subject to a zero borrowing constraint, and earn labor income taxed at the flat rate τ_t . They differ in their level of permanent productivity α_i , and are subject to uninsurable idiosyncratic productivity shocks e_t . They discount the future with the factor β , face a constant death probability ζ , and derive utility from consumption c_t and from their wealth holdings a_t . Their maximization problem is

$$\begin{aligned} V_t(a_{t-1}, e_t, \alpha_i) &= \max_{a_t} \left\{ u(c_t) + v(a_t) + (1 - \zeta)\beta \mathbb{E}_e [V_{t+1}(a_t, e_{t+1}, \alpha_i)] \right\}, \\ \text{s.t. } c_t + a_t &= \underbrace{(1 + r_t)a_{t-1}}_{\text{financial income}} + \underbrace{s(a_{t-1}, A_{t-1})cap_t}_{\text{capital gains}} + \underbrace{\alpha_i e_t Z_t}_{\text{post-tax labor income}} + \underbrace{\chi_t}_{\text{transfers}}, \quad a_t \geq 0. \end{aligned} \quad (2)$$

In Equation (3), r_t is the real return on savings between $t - 1$ and t , known at $t - 1$: it is the ex-ante real rate set one period earlier, so the date-0 return is predetermined when a shock arrives. $Z_t \equiv (1 - \tau_t) w N_t$ is aggregate post-tax labor income, of which the household receives its productivity share $\alpha_i e_t$; N_t denotes the hours worked (set by the union, described below) and w the real wage. χ_t is a lump-sum transfer from the government, cap_t denotes the capital gains (to be defined below), A_{t-1} the total wealth in the economy, and $s(a_{t-1}, A_{t-1})$ is the household's exposure to equity, which is assumed to follow an exogenous, wealth-dependent rule:

$$s(a_{t-1}, A_{t-1}) = \frac{\psi_0}{1 + \exp(-\psi_1(\log(a_{t-1}/A_{t-1}) - \psi_2))} a_{t-1}. \quad (4)$$

This reduced-form rule captures the well-documented heterogeneity in portfolio composition along the wealth distribution without requiring a full portfolio choice prob-

⁵An MIT shock is a one-time unexpected shock. From the model's point of view, this shock is a zero-probability event, and the following transition towards the steady state is deterministic and perfectly foreseen.

lem.⁶

All households supply labor with productivity $\alpha_i e_t$, where the idiosyncratic productivity e_t follows a log-AR(1) process with persistence ρ_e and unconditional standard deviation σ_e , discretized using the Rouwenhorst method, and normalized so that $\mathbb{E}[e_t] = 1$. Each permanent type i has a total mass ω_i , with $\sum_i \omega_i = 1$, and the total endowment of labor productivity is normalized to one, $\sum_i \omega_i \alpha_i = 1$. Through the lens of our model, an increase in permanent labor income inequality is a mean-preserving spread in the distribution of $\{\alpha_i\}$ that preserves this normalization.

Every period, a share ξ of households dies and is replaced by newborn households of the same permanent type with zero assets. Estates are distributed to entering newborns in proportion to their permanent productivity: a newborn of type i receives an endowment $\alpha_i \eta_t / \xi$ on top of its first-period income, where η_t is the aggregate flow of accidental bequests

$$\eta_t = \xi \left(A_{t-1}(1 + r_t) + \text{cap}_t \int s(a_{ss}, A_{ss}) d\Phi_{ss} \right), \quad (5)$$

and Φ_t denotes the joint distribution over wealth, idiosyncratic productivity, and permanent productivity. Mortality plays no direct role in the transmission mechanism, but it disciplines the model: it keeps the wealth distribution non-degenerate despite the taste for wealth and pins down the Pareto tail of wealth in Section 3, and the bequest flow that rides on it makes the multiplier operators of Section 4 invertible.

Period utility functions are given by

$$u(c_t) = \log(c_t), \quad v(a_t) = \gamma \frac{(a_t + \zeta)^{1-\Sigma}}{1-\Sigma},$$

where γ is the intensity of the taste for wealth, ζ a Stone-Geary shifter, and Σ the curvature of wealth in utility; wealth is measured in units of steady-state quarterly output.⁷

The taste for wealth is non-homothetic for two reasons. First, we allow the curvature

⁶The equity share is capped at one, and households with no wealth hold no equity. Because wealth enters relative to the aggregate A_{t-1} , the rule shifts with the level of wealth of the economy considered. Appendix F describes the calibration of (ψ_0, ψ_1, ψ_2) to the Survey of Consumer Finances, following Auclert & Rognlie (2020).

⁷The preference is scale-free in trend output: along a balanced-growth path, ζ and γ scale with trend output as ζY_t and $\gamma Y_t^{\Sigma-1}$, and the detrended problem is stationary.

on wealth Σ to differ from the unit curvature of log consumption, and assume $\Sigma < 1$, so that wealth is a luxury good and the marginal propensity to save increases with permanent income. Second, the Stone-Geary shifter ζ lowers the marginal utility of wealth for poorer households. We explore the implications of this assumption further in Section 3 and Section 5.

2.2 Firms

The supply side is composed of two sectors. A competitive final good producer combines a continuum of imperfectly substitutable intermediate inputs j ,

$$Y_t = \left(\int_j y_{jt}^{\frac{\varepsilon-1}{\varepsilon}} dj \right)^{\frac{\varepsilon}{\varepsilon-1}},$$

while intermediate firms produce using labor, $y_{jt} = l_{jt}$, and set flexible prices. In the symmetric equilibrium, $y_{jt} = Y_t$ and $l_{jt} = L_t$, each intermediate firm charges the constant markup $\mu = \frac{\varepsilon}{\varepsilon-1}$ over its marginal cost, the real wage is constant at $w = \frac{1}{\mu}$, and dividends are a constant share of output:

$$d_t = Y_t - wN_t = Y_t \frac{\mu - 1}{\mu}.$$

The value of the firm p_t is given by the no-arbitrage condition between equity and bonds, where dividends are paid net of the tax τ :

$$\frac{p_{t+1} + (1 - \tau)d_{t+1}}{p_t} = 1 + r_{t+1}. \quad (6)$$

2.3 Unions

A union sets a common level of hours N_t for all households, so that effective demand determines labor demand one for one. Nominal wages are sticky: the union adjusts them subject to a quadratic cost, which yields a standard wage Phillips curve (Auclert et al. 2024)

$$\pi_t^w = \kappa \left(\vartheta N_t^{1/\nu} - (1 - \tau_t) w C_t^{-1} \right) + (1 - \xi) \beta \pi_{t+1}^w, \quad (7)$$

where π_t^w is wage inflation, κ the slope of the Phillips curve, ν the Frisch elasticity, and ϑ a scaling parameter on the disutility of labor, set so that the term in parentheses is zero in steady state. Since prices are flexible and productivity is constant, price inflation equals wage inflation. We set κ close to zero, so that nominal wages are nearly rigid and output is demand-determined.

2.4 Government

The fiscal authority taxes labor income and dividends at the flat rate τ_t , so that tax receipts are

$$T_t = \tau_t (wN_t + d_t) = \tau_t Y_t,$$

issues real one-period bonds B_t , spends G_t , and pays lump-sum transfers χ_t . The budget constraint writes

$$B_t = (1 + r_t)B_{t-1} + G_t + \chi_t - T_t. \quad (8)$$

Importantly, the three instruments differ in their incidence: transfers pay every household the same amount; a tax cut raises post-tax labor income in proportion to productivity, so its dollars accrue mostly to households with high permanent income; and spending enters the goods market directly, without passing through household budgets. Throughout the paper, we therefore distinguish the instrument that pays out a fiscal impulse from the instrument that finances it, and we specify for every experiment, fiscal or monetary, which instrument clears the budget.

Fiscal policy shock. A fiscal shock consists of a path for public debt $\{B_t\}$ and a payout: an increase in spending G_t , an increase in transfers χ_t , or a cut in tax receipts T_t , implemented through the tax rate τ_t . The financing instrument, either transfers or the tax rate, then adjusts residually so that the budget constraint (8) holds in every period. Crossing the three payouts with the two financing instruments yields six packages that can share the same debt path yet differ in their benefit and financing incidence: the same transfer payout can be repaid through future transfer cuts or through future tax increases. A balanced-budget package is the special case $B_t = B_{ss}$.

Monetary policy shock. A monetary shock carries no fiscal impulse, yet it still moves the

budget: a change in r_t changes the cost of servicing the debt, and receipts $\tau_t Y_t$ move with output. Some instrument must adjust. We close the budget with a feedback rule in which a single instrument reacts to the level of public debt,

$$\chi_t = \chi_{ss} - \varphi(B_{t-1} - B_{ss}), \quad G_t = G_{ss} - \varphi(B_{t-1} - B_{ss}), \quad T_t = T_{ss} + \varphi(B_{t-1} - B_{ss}), \quad (9)$$

where φ is the speed of fiscal adjustment and $\chi_{ss} = 0$. Only one rule is active at a time, and the other two instruments stay at their steady-state values: debt absorbs the fiscal space created by the shock, and the active instrument hands it back to households, or spends it, as debt returns to steady state. The choice among the three rules is the monetary counterpart of the financing choice above: a rate cut creates a fiscal windfall, and the rule decides at which incidence it re-enters the economy. Whenever the tax rate is not the active instrument, it is constant at its steady-state value, but tax receipts $T_t = \tau Y_t$ still move with dividends and labor income.

2.5 Central bank

The monetary authority follows a real rate rule, keeping the *ex ante* real interest rate at its steady-state level absent shocks,

$$r_{t+1} = r_{ss} + \varepsilon_t^r, \quad (10)$$

where ε_t^r is an exogenous policy shock. This is equivalent to assuming that the central bank offsets expected inflation one for one in its nominal rule. Under this rule, inflation is irrelevant for all real variables in the model (Auclert et al. 2024), and the wage Phillips curve (7) only determines the path of inflation.

2.6 Mutual fund

A mutual fund receives all savings from households and invests them in the two assets available in the economy, government bonds B_t and firm equity p_t , so that $A_t = p_t + B_t$. The fund is competitive and makes zero profits. In steady state, the no-arbitrage condition (6) ensures that the return on equity equals the return on bonds, so households earn r_t on their savings and $cap_t = 0$.

However, at $t = 0$ after an unexpected shock, the no-arbitrage condition does not hold: the jump in p_0 and d_0 was not priced into $p_{-1} = p_{ss}$. The unexpected excess return (or underperformance) on equity is paid to households according to the portfolio rule of Equation (4), and the balance sheet of the mutual fund at $t = 0$ writes

$$A_{ss}(1 + r_{ss}) + cap_0 \int s(a_{ss}, A_{ss}) d\Phi_{ss} = p_0 + (1 - \tau)d_0 + (1 + r_{ss})B_{ss}, \quad (11)$$

which implies that capital gains in period 0 are equal to

$$cap_0 = \frac{p_0 - p_{ss} + (1 - \tau)(d_0 - d_{ss})}{\int s(a_{ss}, A_{ss}) d\Phi_{ss}}.$$

2.7 Market clearing and equilibrium

Aggregate consumption and savings are obtained by integrating the policy functions over the distribution Φ_t :

$$C_t = \int c_t d\Phi_t, \quad A_t = \int a_t d\Phi_t.$$

Definition 1 (Equilibrium). *A competitive equilibrium is defined as paths for household decisions $\{a_t, c_t\}_{t \geq 0}$, prices $\{r_t, p_t\}_{t \geq 0}$ and the date-0 capital gain cap_0 , fiscal policy $\{\tau_t, \chi_t, G_t, B_t\}$, aggregate quantities $\{Y_t, N_t, d_t, \eta_t\}_{t \geq 0}$, and distributions $\{\Phi_t\}_{t \geq 0}$, such that:*

1. *Households solve their maximization problem given prices and hours;*
2. *The value of the firm satisfies the no-arbitrage condition (6);*
3. *The union sets wages according to Equation (7);*
4. *The central bank follows the rule (10) and the government budget constraint (8) and its fiscal rule hold;*
5. *The mutual fund balance sheet holds;*
6. *The sequence of distributions $\{\Phi_t\}$ is consistent with the policy functions;*
7. *Markets clear:*

(a) *The asset market clears: $A_t = p_t + B_t, \quad \forall t,$*

(b) *The goods market clears:* $Y_t = C_t + G_t, \quad \forall t,$

(c) *The labor market clears:* $L_t = N_t, \quad \forall t.$

3 Calibration and Empirical Fit

We calibrate the model at quarterly frequency to the United States in 2019.⁸ The strategy has three blocks. First, standard parameters, the income process, and the fiscal block are set outside the model. Second, the curvature of the taste for wealth Σ and the death probability ξ are pinned down by two analytical restrictions on the Pareto tails of wealth and consumption, following Gaillard et al. (2023). Third, the two remaining preference parameters (γ, ζ) are chosen by least squares to match the profile of hand-to-mouth shares across permanent labor income quintiles in the PSID, together with the top-10% wealth share, and the discount factor β is set so that the asset market clears. Table 1 reports all parameter values.

3.1 Calibration Strategy

Taste for wealth and death probability. Following Gaillard et al. (2023), we derive an analytical approximation of the Pareto tails of wealth and consumption in the model, which yields

$$\zeta^a = \frac{-\log(1 - \xi)}{\log(1 + r)}, \quad \zeta^c = \frac{\zeta^a}{\Sigma}.$$

Appendix E provides the derivation. Targeting a wealth tail of $\zeta^a = 1.48$ and a consumption tail of $\zeta^c = 3.11$, with the restriction evaluated at the quarterly real rate of 0.5% (2% annualized), gives $\Sigma = 0.48$ and $\xi = 0.0074$.⁹

Given (Σ, ξ) , we choose the strength of the taste for wealth γ and the Stone–Geary shifter ζ (unrelated to the tail coefficients ζ^a, ζ^c) by least squares on six moments: the five hand-to-mouth shares by permanent labor income quintile measured in the 2019 PSID wave (income year 2018), and the top-10% wealth share in the 2018 DINA series (69%). In the model, we classify a household as hand-to-mouth when its net worth is below two months of its labor income, the same threshold as in Section 1; since the model has a single asset, we compare this share to the sum of the poor and wealthy hand-to-mouth shares in the data. The fit yields $\gamma = 0.054$ and $\zeta = 70.3$, so that

⁸We label the calibration 2019, the year of the latest PSID wave we use. The 2019 wave records income year 2018, and we use the 2018 DINA distributions accordingly; only the SCF portfolio fit uses 2019 data. The 2018 and 2019 DINA shares are nearly identical.

⁹This implies an annual death probability of roughly 2.9%, and thus an expected (working) lifetime of about 34 years.

the Stone-Geary shift is about 3.5 times steady-state aggregate wealth. Finally, the discount factor $\beta = 0.990$ clears the asset market at the 2% annual rate.

Labor income. Permanent labor income levels $\{\alpha_i\}$ match the labor income shares of seven groups (the four bottom quintiles, P80–90, P90–95, and the top 5%) computed from the DINA microdata of Piketty et al. (2018) for 2018. Idiosyncratic productivity e_t follows an AR(1) in logs with annual persistence $\rho_e = 0.91$ and unconditional standard deviation $\sigma_e = 0.39$, estimated by minimum distance on the autocovariance structure of residual log earnings in our PSID sample (Appendix D).

Government and firms. The rest of our calibration follows Auclert et al. (2025) closely. We normalize steady-state quarterly output to $Y_{ss} = 1$ and set public debt to $B_{ss} = 4$, an annual debt-to-output ratio of 100%. Government spending is $G_{ss} = 0.20$, and the tax rate $\tau_{ss} = 0.22$ is the residual that closes the government budget constraint. The baseline has no tax progressivity, and the fiscal-rule coefficient is $\varphi = 0.05$; Section 5 varies both the coefficient and the instrument of the rule. The markup $\mu = 1.11$ is chosen so that total wealth, $p_{ss} + B_{ss}$, matches an annual wealth-to-output ratio of 5.¹⁰

Portfolio shares. The parameters of the portfolio rule (4), (ψ_0, ψ_1, ψ_2) , are fitted to the equity and business wealth shares observed in the 2019 Survey of Consumer Finances, following Auclert & Rognlie (2020). Appendix F describes the fit.

3.2 Model Fit

Figure 3 compares hand-to-mouth shares by permanent labor income quintile in the model and in the data. The model reproduces the steep gradient documented in Section 1: the hand-to-mouth share falls from 56% in the bottom quintile to 12% in the top quintile, against 64% and 22% in the data, and the aggregate share is 39% against 41%.¹¹ The model overstates the decline at the very top, where the wealthy hand-to-mouth, absent from a one-asset model, account for most of the remaining incidence in the data.

¹⁰The markup determines steady-state profits and therefore the equity value, $p_{ss} = (1 - \tau)d_{ss}/r_{ss}$.

¹¹The calibration targets the 2019-wave shares; the figure reports the pooled 1999–2019 shares, which are close.

Table 1: Calibration of the HANK model

Parameter	Description	Value	Source
<i>Households</i>			
β	Discount factor	0.990	Asset market clearing at annual r of 2%.
Σ	Curvature on wealth	0.48	Pareto coeff. on wealth/cons. tails (1.48/3.11).
γ	Strength of taste for wealth	0.05	HtM share by PLI quintile (PSID, 2018).
ζ	Stone-Geary shifter	70.29	HtM share by PLI quintile (PSID, 2018).
ξ	Death probability (quarterly)	0.0074	Wealth Pareto tail 1.48.
ρ^e	Persistence of income shocks (annual)	0.91	PSID, minimum-distance estimation.
σ^e	Uncond. std of persistent component	0.39	PSID, minimum-distance estimation.
$\{\psi_0, \psi_1, \psi_2\}$	Portfolio shares	(3.62, 0.48, 6.91)	SCF (2019).
<i>Government and Firms</i>			
B_{ss}	Public debt	4.00	Annual B/Y of 100%.
G_{ss}	Gov. spending	0.20	National accounts.
τ_{ss}	Tax rate	0.22	Residual (gov. budget).
ϕ	Tax progressivity (HSV)	0.00	Baseline: none.
φ	Fiscal rule parameter	0.05	Standard; swept in Section 5.
μ	Markup	1.11	Annual W/Y of 5.0.

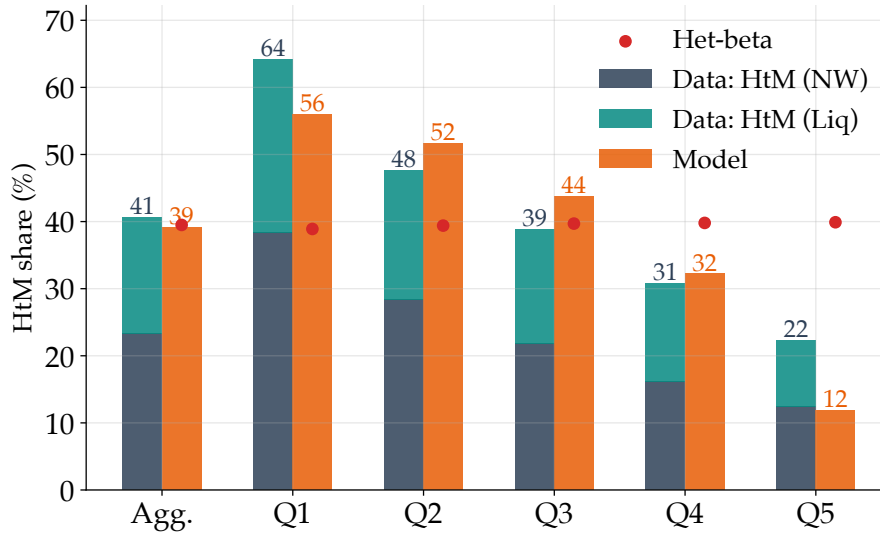


Figure 3: Hand-to-Mouth Shares by Permanent Labor Income Quintile

Note: Bars show hand-to-mouth shares in the PSID (pooled 1999–2019 sample; net-worth and wealthy hand-to-mouth stacked) and in the model, in the aggregate and by PLI quintile. Dots show the homothetic benchmark with heterogeneous discount factors described in Section 3.4.

Table 2 compares the model wealth distribution with the 2018 DINA series of Piketty

et al. (2018). The top-1% wealth share, which is untargeted, comes out at 34%, as in the data, and the bottom half of the distribution holds essentially no wealth, as in the data. Two discrepancies remain. First, the model understates the middle 40% (16% against 30%) and correspondingly overstates the top 10% (84% against 69%, despite the top-10% share being targeted): with two parameters against six moments, the least-squares fit favors the hand-to-mouth profile, and the “missing middle” typical of one-asset models persists. Second, the model understates the extreme tail (8% against 18% for the top 0.1%). We return to both points below.

Table 2: Wealth distribution for the year 2018

	Wealth shares (%)					Annual MPC (%)		
	Bottom 50%	Middle 40%	Top 10%	Top 1%	Top 0.1%	Transfer	Income-wtd.	HtM (%)
Data	1	30	69	34	18			41.2
Model	0	16	84	34	8	36.4	17.3	39.1

The last columns of Table 2 report the aggregate annual marginal propensity to consume out of a lump-sum transfer, 0.36 in the model. This moment is untargeted and sits within the range of empirical estimates: matching the concentration of hand-to-mouth status among low permanent income households delivers a high aggregate MPC without any discount-factor heterogeneity.

Figure 4 plots the counter-cumulative distribution of wealth in the model against a Pareto tail with the empirical coefficient $\zeta^a = 1.48$. The analytical approximation used in the calibration captures the model’s tail behavior well: the model reproduces the observed slope and generates a significant mass of households with wealth above a billion dollars. The homothetic benchmark, discussed in Section 3.4, generates a much thinner tail.

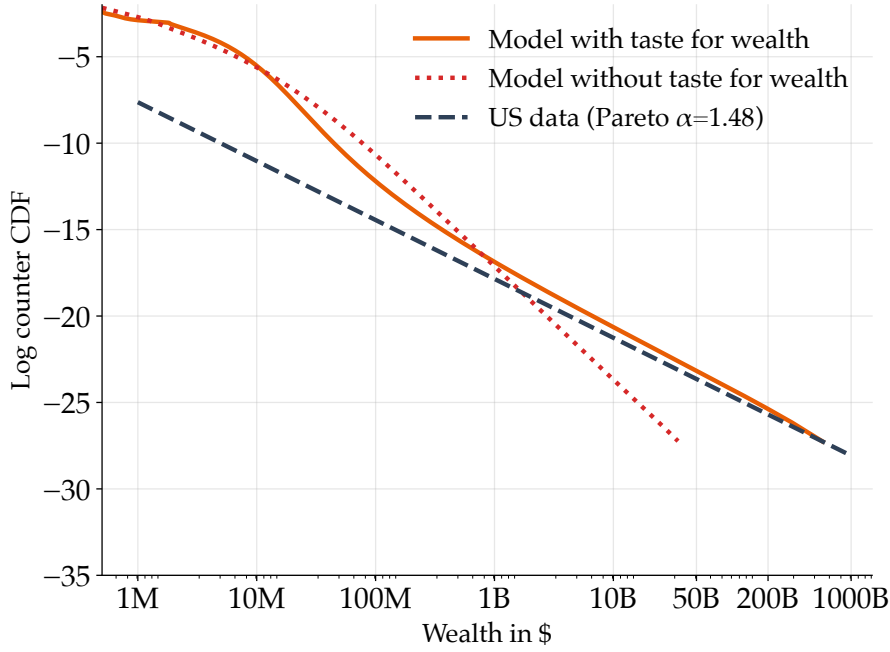


Figure 4: Wealth Pareto Tail

Note: Log counter-cumulative distribution of wealth, expressed in dollars. The two model lines are the baseline (labeled “with taste for wealth”) and the het-beta benchmark of Section 3.4 (labeled “without taste for wealth”). The dashed line shows a Pareto tail with the empirical coefficient $\zeta^a = 1.48$ (α in the figure legend), anchored at the end of the baseline model’s tail.

3.3 How a non-homothetic taste for wealth matches both the wealth distribution and the HtM–PLI gradient

It is well known that standard HANK models with only idiosyncratic shocks generate too little wealth inequality and too low a share of hand-to-mouth households when calibrated to match the observed wealth-to-output ratio (Kaplan et al. 2018, Auclert et al. 2024). This, in turn, implies an aggregate MPC well below empirical estimates. In these environments, idiosyncratic risk and precautionary saving are the only sources of heterogeneity: households accumulate buffers up to a target level of wealth and then dissave at a rate that prevents the emergence of very rich households. These models thus cannot reproduce the high saving rates observed at the top of the wealth distribution. When the calibration is adjusted to match the aggregate wealth-to-output ratio, too much wealth is assigned to the lower and middle parts of the distribution, mechanically lowering their MPCs. The failure to match high saving rates at the top and a high aggregate MPC are two sides of the same issue.

Introducing a taste for wealth resolves this tension by providing an additional saving

motive that is strongly increasing in wealth. Because households derive direct utility from wealth, and because this motive is non-homothetic, the very rich behave as if wealth were a luxury good: once their consumption needs are met, they devote almost all marginal income to additional saving. This mechanism allows the model to match the high saving rates documented for the top of the distribution (Dynan et al. 2004, Kumhof et al. 2015, Straub 2019).

Figure 5 shows the policy functions for consumption and next-period assets at the top of the wealth distribution. As wealth increases, savings approach the simple accumulation rule $a_t \approx (1 + r)a_{t-1}$: wealthy households save nearly all of their disposable resources, and their behavior converges to the analytical asymptote derived in Appendix E. The death probability keeps the distribution stationary, as it prevents indefinite wealth accumulation by rich households.

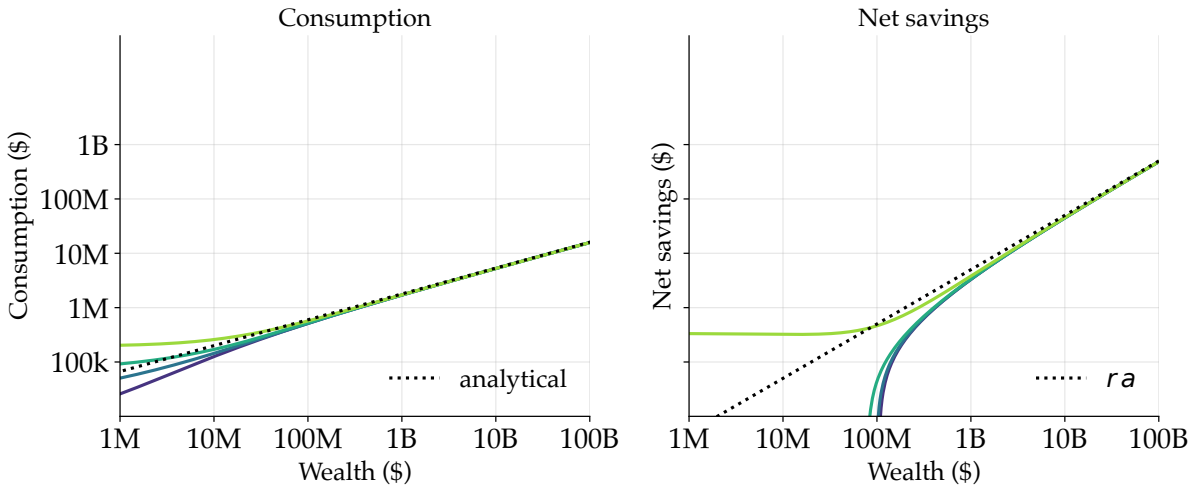


Figure 5: Savings and Consumption Behavior of Wealthy Households

Note: This figure plots the numerical policy functions for net savings and consumption as functions of wealth, expressed in dollars (x-axis), for different idiosyncratic and permanent types (colored lines). Dotted black lines plot the analytical approximation of those policy functions as wealth $\rightarrow \infty$, derived in Appendix E.

Households at the bottom of the wealth distribution behave differently. Because the taste for wealth is negligible at low wealth levels, their decisions resemble those of standard buffer-stock agents, driven primarily by precautionary motives. The strong saving of wealthy households pushes the equilibrium return to a level such that $\beta(1 + r) \ll 1$, which weakens the incentive of low-wealth households to save and keeps them near the borrowing constraint. Because low permanent income households expect their income to remain low, they hold small buffers relative to high permanent

income households, and the model concentrates hand-to-mouth status at the bottom of the permanent income distribution, as in Figure 3.

3.4 A homothetic benchmark: heterogeneous discount factors

To isolate the role of the non-homothetic taste for wealth, we build a homothetic benchmark of the baseline model with heterogeneous discount factors, *het-beta* for short. We set $\gamma = 0$ and instead let the discount factor take two values, $\beta_j \in \{\beta_L, \beta_H\}$, following a two-state Markov chain, the standard device to generate wealth concentration and high MPCs in heterogeneous-agent models (Krusell & Smith 1998, Auclert et al. 2025). We calibrate its three parameters, the patient share $\omega = 0.15$, the spread $\beta_H - \beta_L = 0.022$, and the quarterly switching probability $q_\beta = 0.005$, so that *het-beta* reproduces the baseline model's aggregate hand-to-mouth share and top-10% and top-1% wealth shares; the mean discount factor again clears the asset market.

By construction, the two models share the same aggregate wealth concentration and hand-to-mouth incidence, and their aggregate annual MPCs are nearly identical (0.36 in both). They differ in how these features correlate with permanent income. In *het-beta*, hand-to-mouth status is driven by the discount-factor type, which is independent of permanent income: the hand-to-mouth profile across permanent income quintiles is flat, between 39% and 40% (dots in Figure 3), at odds with the data. *Het-beta* also generates a thinner extreme tail than the baseline model (Figure 4), since patient households eventually stop accumulating once their buffer motive is satiated. This benchmark is the relevant comparison for everything that follows: under homothetic preferences, the distribution of permanent labor income is close to neutral for aggregate behavior, and the differences between the two models identify the role of the taste for wealth.

4 Stabilization Policy with a Taste for Wealth

This section studies how fiscal and monetary policy differ in our non-homothetic model compared to the *het-beta* benchmark of Section 3.4. The two economies share nearly the same aggregate hand-to-mouth share, top-10% and top-1% wealth shares,

and aggregate annual MPC (0.36 in both); they differ only in the joint distribution of MPCs and permanent labor income, the steep gradient of Figure 3 against a flat profile. This single difference has three consequences.

First, because hand-to-mouth status is concentrated among low-PLI households in the baseline, while it is unrelated to PLI in het-beta, the share of aggregate labor income earned by HtM households is lower in the baseline (Section 4.1). The gap between the MPC out of a transfer and the MPC out of labor income is therefore larger in the baseline than in het-beta. Second, this gap implies that the incidence of fiscal policy matters more in the baseline model (Section 4.2): who receives a package's payouts and who repays them becomes a first-order determinant of the multiplier, to the point that the same deficit path can stimulate or contract output depending on its composition. Third, the same logic carries to monetary policy (Section 4.3): a rate cut creates fiscal space, and the output response depends much more on how the government rebates it in the baseline than in het-beta, while the direct rate channel is nearly identical across the two models.

4.1 The aggregate consumption function with non-neutral permanent income

In our model, household decisions depend on aggregates only through the path of aggregate post-tax labor income $Z_t \equiv (1 - \tau_t)wN_t$, of which household (a, e, i) receives the share $\alpha_i e_t$; the path of transfers χ_t , received one for one; the path of real returns r_t ; and the aggregate date-0 capital gain cap , received in proportion to equity exposure, $s(a_{ss}, A_{ss})/\bar{S}$ with $\bar{S} \equiv \int s(a_{ss}, A_{ss}) d\Phi_{ss}$.¹² Writing aggregate consumption as $C_t = \mathcal{C}_t(\{Z_s, \chi_s, r_s\}_{s \geq 0}, cap)$ ¹³ and totally differentiating,

$$d\mathbf{C} = \mathbf{M}^Z d\mathbf{Z} + \mathbf{M}^\chi d\chi + \mathbf{M}^r dr + \mathbf{m}^{cap} dcap, \quad (12)$$

¹²Aggregate consumption also depends on the path of bequest flows η_t . We omit this flow to save on notation; the quantitative model keeps it, and all figures and tables are computed from the full model. The term is proportional to the quarterly death rate ζ : restoring it makes Equation (17) reproduce the full model to numerical precision.

¹³Bold letters denote sequences, e.g. $d\mathbf{Y} = (dY_0, dY_1, \dots)'$, and matrices operating on them; $\mathbf{q} \equiv (1, (1+r)^{-1}, (1+r)^{-2}, \dots)'$ is the steady-state discount vector, $\mathbf{L}$ the lag matrix, and $\mathbf{I}$ the identity. Following Auclert et al. (2024) and Auclert et al. (2025), we collapse the household block into a set of Jacobians; Appendix A contains all derivations and proofs.

where $\mathbf{M}_{t,s}^Z \equiv \partial \mathcal{C}_t / \partial Z_s$, $\mathbf{M}_{t,s}^\chi \equiv \partial \mathcal{C}_t / \partial \chi_s$, $\mathbf{M}_{t,s}^r \equiv \partial \mathcal{C}_t / \partial r_s$ and $\mathbf{m}_t^{cap} \equiv \partial \mathcal{C}_t / \partial cap$, are the sequence-space Jacobians of the household block: entry (t, s) is the change in aggregate consumption at time t caused by a date- s change in the corresponding input. These four matrices are the only place the two models can differ: every difference between the non-homothetic model and the benchmark het-beta model can be traced back to differences in $\mathbf{M}^Z$, $\mathbf{M}^\chi$, $\mathbf{M}^r$, and $\mathbf{m}^{cap}$.

Figure 6 compares these Jacobians across the two models. The transfer Jacobians are similar: both models spend a flat dollar in nearly the same way (panel a). The interest-rate Jacobians are close as well, and nearly identical in present value (panel c), a point Section 4.3 returns to. The difference is concentrated in the labor-income Jacobian: on impact, a labor-income dollar raises consumption by 0.12 in the baseline against 0.22 in het-beta (panel b), and cumulating the discounted responses, it generates 20% less spending in present value than a transfer dollar, against 2% in het-beta. In the baseline, a dollar earned as labor income is spent less, and later, than the same dollar received as a transfer; het-beta's small gap is a timing gap that washes out in present value.

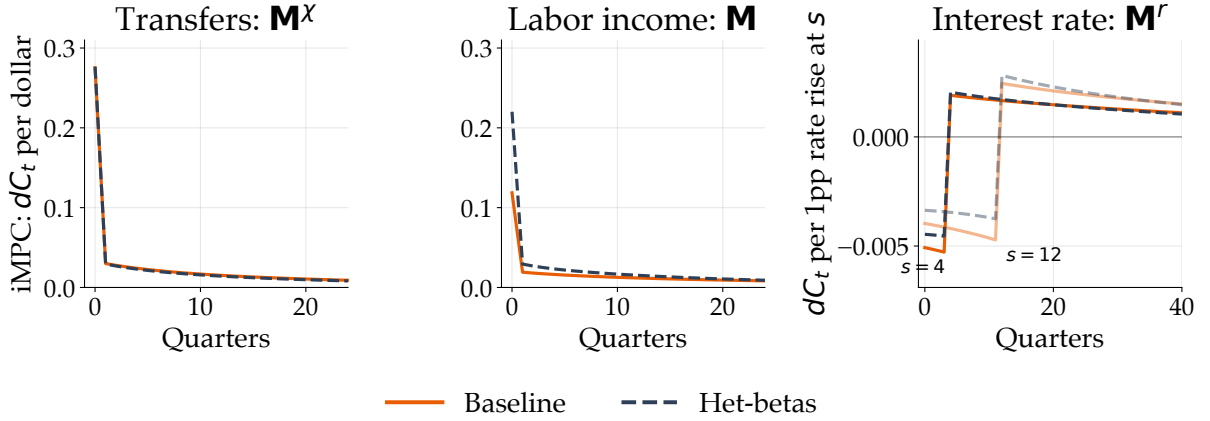


Figure 6: The consumption Jacobians in the baseline and het-beta models

Note: Consumption responses of the household block to a small shock of 0.1% of quarterly GDP. Panels (a) and (b): path of dC_t per dollar of a date-0 payout, distributed equally across households (panel a, the first column of $\mathbf{M}^\chi$) or in proportion to labor income (panel b, the first column of $\mathbf{M}^Z$). Panel (c): dC_t per 1 percentage point rise in the quarterly real rate at $s = 4$ and $s = 12$ (columns of $\mathbf{M}^r$). High-inequality (2019) calibration.

Why does the labor-income Jacobian differ across models when the transfer Jacobian does not? Because in the baseline, MPCs are correlated with permanent labor income. To state this, note that a marginal dollar at date s raises the consumption of household (a, e, i) at date t by the same amount $m_{t,s}(a, e, i)$ whether the dollar is labeled labor in-

come, transfer, or capital gain; the aggregate Jacobians differ only in how they weight households. Let $\bar{m}_{i,t,s} \equiv \mathbb{E}_i[m_{t,s}]$ denote the average kernel of permanent type i .

Proposition 1 (Incidence and iMPCs). *Let dX_s be an aggregate payout at date s distributed with incidence $\iota_s(a, e, i)$, normalized so that $\mathbb{E}[\iota_s] = 1$. Then*

$$\frac{\partial \mathcal{C}_t}{\partial X_s} = \mathbb{E}[\iota_s m_{t,s}] = \mathbf{M}_{t,s}^\chi + \text{Cov}(\iota_s, m_{t,s}).$$

In particular, for labor income ($\iota_s = \alpha_i e_s$) and for capital gains ($\iota_0 = s(a_{ss}, A_{ss})/\bar{S}$),

$$\mathbf{M}_{t,s}^Z = \mathbf{M}_{t,s}^\chi + \underbrace{\sum_i \omega_i (\alpha_i - 1) \bar{m}_{i,t,s}}_{\text{iMPC-PLI covariance}} + \underbrace{\sum_i \omega_i \alpha_i \text{Cov}_i(m_{t,s}, e_s)}_{\text{iMPC-transitory covariance}}, \quad (13)$$

$$\mathbf{m}_t^{\text{cap}} = \mathbf{M}_{t,0}^\chi + \text{Cov}(s(a_{ss}, A_{ss})/\bar{S}, m_{t,0}). \quad (14)$$

Proof. Appendix A. □

Proposition 1 says that the difference between any two income Jacobians is a covariance between who receives the dollar and who spends it. For labor income the covariance has two parts: a between-type part, since low-PLI households have high iMPCs but a below-average income share α_i , and a within-type part driven by transitory shocks. For capital gains, Equation (14) says that $\mathbf{m}^{\text{cap}}$ falls short of $\mathbf{M}^\chi$ whenever equity is concentrated among low-MPC wealthy households; this covariance prices the valuation channel of monetary policy in Section 4.3.

Under homothetic preferences the consumption policy is linear in permanent income, so every type spends a marginal dollar the same way: the iMPC-PLI covariance in Equation (13) is identically zero, and the remaining wedge, the transitory covariance, does not depend on the PLI distribution (Lemma 2 in Appendix A). This covers hetero-beta, whose discount-factor heterogeneity leaves preferences homothetic. It is a statement about the wedge only; the stronger result that *all* Jacobians are invariant to the PLI distribution is Proposition 4 in Appendix A, put to work in Section 5.

Figure 7 shows why the between-type covariance is large in the baseline. Solving the

household problem over a continuum of permanent income levels, the annual MPC is a steeply decreasing, convex function of permanent income, falling from above 0.4 at the bottom to below 0.05 at the top, while the EIS rises from about 0.75 to 0.95, and the hand-to-mouth share falls in step (panel b). In the homothetic het-beta benchmark, all three profiles are flat. The taste for wealth generates the slopes: wealth is a luxury good, so high-permanent-income households put marginal income into an asset they value for its own sake, while low-permanent-income households, facing $\beta(1+r) \ll 1$, stay near the borrowing constraint.

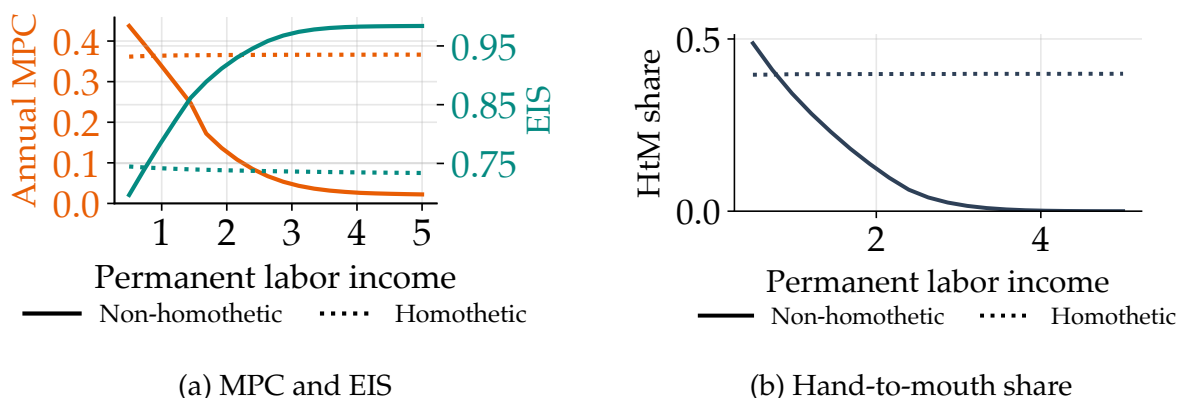


Figure 7: MPC, EIS, and hand-to-mouth share as functions of permanent labor income

Note: Annual MPC out of a transfer (panel a, left axis), EIS (panel a, right axis), and hand-to-mouth share (panel b) as functions of permanent labor income, in the baseline (full lines) and in the homothetic het-beta benchmark of Section 3.4 (dotted lines). To compute these statistics, we keep prices constant, solve the household problem for different values of permanent labor income, and aggregate over the distribution of wealth and idiosyncratic shocks.

The rest of this section traces the gap between $\mathbf{M}^X$ and $\mathbf{M}^Z$ through the equilibrium, first for fiscal policy, then for monetary policy.

4.2 Implications for fiscal policy

Because transfers, taxes, and purchases reach different households, the incidence of a fiscal package, who receives its payouts and who repays them, becomes more important in the baseline non-homothetic model compared to the benchmark het-beta model. To derive the importance of this heterogeneous incidence, substitute the identities tying $d\mathbf{Z}$ and $d\text{cap}$ to output and taxes into Equation (12) and impose goods market clearing $d\mathbf{Y} = d\mathbf{C} + d\mathbf{G}$ (Appendix A.1). This delivers the intertemporal Keynesian

cross of our model,

$$d\mathbf{Y} = \mathbf{M}^T (d\mathbf{Y} - d\mathbf{T}) + \mathbf{M}^\chi d\chi + \overline{\mathbf{M}}^r d\mathbf{r} + d\mathbf{G}, \quad (15)$$

where $\mathbf{M}^T \equiv \frac{1}{\mu}\mathbf{M}^Z + \frac{\mu-1}{\mu}\mathbf{m}^{cap}\mathbf{q}'$ blends the labor-income and dividend-valuation responses to post-tax income $d\mathbf{Y} - d\mathbf{T}$, and $\overline{\mathbf{M}}^r \equiv \mathbf{M}^r - p\mathbf{m}^{cap}\mathbf{q}'$ nets the valuation loss from higher rates out of the direct rate response. By Proposition 1, $\mathbf{M}^T$ inherits the wedge on both margins: labor income through the iMPC–PLI covariance of Equation (13), capital gains through the equity-concentration covariance of Equation (14).

The change in tax receipts $d\mathbf{T} = \tau d\mathbf{Y} + Y d\tau$ is tied to the policy instruments by the government budget constraint (8). Totally differentiating it,

$$d\mathbf{T} = d\mathbf{G} + d\chi + B d\mathbf{r} + [(1+r)\mathbf{L} - \mathbf{I}] d\mathbf{B}. \quad (16)$$

Fiscal policy announces paths for its instruments and for debt, and one instrument adjusts residually so that Equation (16) holds period by period. The debt term is present-value neutral, $\mathbf{q}'[\mathbf{I} - (1+r)\mathbf{L}] d\mathbf{B} = 0$, by telescoping.¹⁴

The output response to any policy package then follows from a single formula. Write the path of each fiscal instrument as a payout, $d\epsilon^\chi \equiv d\chi$, $d\epsilon^G \equiv d\mathbf{G}$, and $d\epsilon^T \equiv -Y d\tau$ (positive for a tax cut). Each instrument has its own benefit incidence, priced by its Jacobian: transfers reach all households equally ($\mathbf{M}^\chi$), tax changes follow post-tax income ($\mathbf{M}^T$), and purchases are spent one for one ($\mathbf{M}^G \equiv \mathbf{I}$).

Proposition 2 (Fiscal policy incidence). *Let $x \in \{\chi, T, G\}$ denote the type of fiscal payout and $f \in \{\chi, T, G\}$ the instrument used for its financing. For a given debt path $d\mathbf{B}$, under the real-rate rule ($d\mathbf{r} = 0$), the output response solves*

$$\left[\mathbf{I} - \mathbf{M}^T - \tau(\mathbf{M}^f - \mathbf{M}^T) \right] d\mathbf{Y} = \underbrace{\mathbf{M}^f [\mathbf{I} - (1+r)\mathbf{L}] d\mathbf{B}}_{\text{Deficit-timing effect}} + \underbrace{(\mathbf{M}^x - \mathbf{M}^f) d\epsilon^x}_{\text{Heterogeneous incidence effect}}. \quad (17)$$

Proof. Appendix A. □

¹⁴The multiplier operators below are invertible in our model, in contrast to the benchmark of Auclert et al. (2024), where discounted iMPCs sum to one and the operator is singular. The identity fails here because the Jacobians hold the bequest flow received by newborns fixed: the estates of dying households do not re-enter the operator as newborn endowments. Numerically, $\mathbf{q}'\mathbf{M}_{\cdot,0}^\chi = 0.85$; see Appendix A.1.

Equation (17) splits any fiscal package into two effects. The deficit-timing effect is the classical intertemporal Keynesian cross force: a deficit shifts income across time, and because iMPCs are front-loaded, households spend today's dollar faster than they cut spending for the dollar taken back later, so even a present-value-neutral deficit raises output. When the same instrument is used to pay out and to finance, $x = f$, this is the only effect, and Equation (17) reduces to the intertemporal Keynesian cross of Auclert et al. (2024), evaluated at the financing instrument's iMPCs.

The heterogeneous incidence effect appears when $x \neq f$: the households who receive the payout then differ, in type and in time, from those who finance it, and the gap $\mathbf{M}^x - \mathbf{M}^f$ between the two incidences can strengthen or weaken the response to a given package. This effect does not need a deficit: even with $d\mathbf{B} = 0$, a balanced-budget package redistributes income across households and moves output whenever the recipients spend differently from the payers. Incidence also enters the multiplier, through $\tau(\mathbf{M}^f - \mathbf{M}^T)$: the receipts the expansion generates endogenously, $\tau d\mathbf{Y}$, are handed back through the financing instrument.

Figure 8 shows that this heterogeneous incidence matters quantitatively. It reports cumulative multipliers for six deficit-financed packages: a payout through government spending, transfers, or a tax cut, each repaid later through lower transfers or through higher taxes. In the baseline, switching only the repayment instrument moves each multiplier by more than 0.5; in het-beta, by less than 0.25. The incidence effect is large enough to flip the sign of a package: a tax cut repaid by future transfer cuts has a cumulative multiplier of -0.61 , because it hands dollars to low-MPC, high-income households and takes them from high-MPC households. Government spending shows the same reversal relative to its own benchmark: a balanced-budget expansion has a multiplier of one, and repayment through transfer cuts drags the deficit-financed multiplier down to 0.57 , the counterpart of the negative tax-cut multiplier. In het-beta no package crosses these thresholds: the four transfer-tax multipliers sit between 0.16 and 0.67 , and both spending multipliers stay above one.

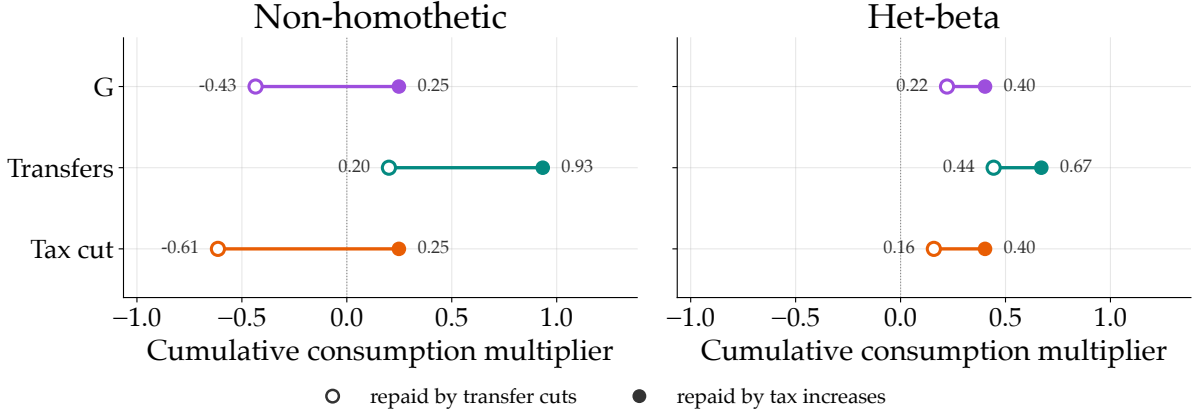


Figure 8: Cumulative multipliers by fiscal package: payout and financing instrument

Note: Cumulative multipliers $(\sum_t Q^t dY_t) / (\sum_t Q^t d\epsilon_t)$, $Q = 1/(1+r)$, for deficit-financed packages combining a payout instrument (government spending, transfers, or a tax cut) with a financing instrument (future transfer cuts, hollow markers, or future tax increases, filled markers). Left panel: baseline model. Right panel: het-beta benchmark. The figure reports consumption multipliers: they equal the output multipliers for the transfer and tax-cut payouts, and the output multiplier minus one for the spending payout.

4.3 Implications for monetary policy

We now turn to monetary policy. As shown in Figure 6, households react to the rate path itself similarly in the two models: the present value of the direct effect of a rate cut on consumption is nearly the same. What differs is what the cut sets in motion. The indirect effect, the spending out of the labor income the expansion generates, is weaker in the baseline, and the fiscal side of the shock matters far more: a rate cut lowers the interest bill on public debt and raises tax receipts, and switching between the two household rebates of this fiscal space moves the output response seven times more in the baseline than in het-beta (98% against 14%).

To see both facts in one equation, note that monetary policy affects fiscal policy by changing the interest rate the government pays on its public debt. Repeating the derivation of Proposition 2 for a rate path $d\mathbf{r}$ (with $dr_0 = 0$ and $d\epsilon^x = 0$), the output response solves

$$[\mathbf{I} - \mathbf{M}^T - \tau(\mathbf{M}^f - \mathbf{M}^T)] d\mathbf{Y} = [\overline{\mathbf{M}}^r - B\mathbf{M}^f] d\mathbf{r} + \mathbf{M}^f [\mathbf{I} - (1+r)\mathbf{L}] d\mathbf{B}. \quad (18)$$

A rate cut moves output through the direct channel $\overline{\mathbf{M}}^r$ and through the public finances, and the debt path $d\mathbf{B}$ is not a free object: it is generated by the fiscal rule (9), with the financing instrument f returning debt to steady state at speed φ . We read

Equation (18) in two steps.

Suppose first that the government has no debt, $B = 0$ and $d\mathbf{B} = 0$, and balances its budget with the tax rate, $f = T$. Equation (18) reduces to $d\mathbf{Y} = [\mathbf{I} - \mathbf{M}^T]^{-1} \overline{\mathbf{M}}^r d\mathbf{r}$. The impulse is nearly the same in the two models: feeding the same rate path through $\mathbf{M}^r$ alone, partial-equilibrium consumption rises by 0.035 in the baseline against 0.029 in het-beta over the first year, and the present-value responses are nearly identical, consistent with the two models' nearly equal aggregate EIS (0.75 against 0.74). Without debt, the only difference between the models is the indirect effect, priced by $\mathbf{M}^T$ inside the operator: in the baseline, the labor income the expansion generates accrues mostly to households who save it (Proposition 1), so the amplification of the same impulse is weaker.

With debt, $B > 0$, the cut also relaxes the government budget: the cost of rolling over the debt falls, the $B d\mathbf{r}$ term, and the fiscal rule hands this fiscal space back to the economy through the residual instrument, higher transfers, higher spending, or lower taxes, letting debt absorb part of the windfall and repaying it at speed φ , the $d\mathbf{B}$ term. The relative incidence of the instruments now matters: $\mathbf{M}^f$ prices the impulse and the multiplier. Since $\mathbf{M}^G = \mathbf{I}$ in both models and $\mathbf{M}^\chi$ is nearly common to the two (panel a), while $\mathbf{M}^T$ sits far below $\mathbf{M}^\chi$ only in the baseline, the choice of instrument matters in the baseline, and the household rebates are close to interchangeable in het-beta.

Figure 9 quantifies this. It reports the cumulative consumption semi-elasticity, the cumulated consumption response per unit of cumulated rate change (Kaplan et al. 2018), to the same expansionary shock of 25 basis points to the quarterly real rate (about 100 basis points annualized; persistence $\rho^r = 0.9$) when the residual instrument of the fiscal rule is transfers (χ), the tax rate (T), or government spending (G). In het-beta, the two household rebates are nearly interchangeable, a semi-elasticity of 11.6 under the transfers closure against 10.2 under the tax closure, and the spending closure, which hands nothing back to households, delivers 9.2. In the baseline, switching the rebate from taxes to transfers moves the semi-elasticity from 5.2 to 10.4, a 98% difference, and the tax closure is barely distinguishable from the spending closure, 4.9: rebating the windfall through the tax rate raises consumption hardly more than not rebating it to

households at all.¹⁵ On impact the closures are nearly indistinguishable in both models: the closure margin operates through the accumulated fiscal flows, not the initial demand impulse.



Figure 9: Consumption response to a monetary easing by fiscal rule

Note: Cumulative consumption semi-elasticity to the same expansionary shock of 25 quarterly basis points ($\rho^r = 0.9$): the discounted cumulated consumption response, $\sum_t (1+r)^{-t} dC_t/C$, per unit of discounted cumulated change in the real rate, $-\sum_t (1+r)^{-t} dr_t/(1+r)$. The financing instrument follows the feedback rule (9) at the baseline speed $\varphi = 0.05$. High-inequality (2019) calibration.

The intuition is the same as for fiscal policy: the closure decides at which incidence the fiscal windfall re-enters household budgets. In the baseline, rebating it through flat transfers targets the high-MPC, low-PLI households, while rebating it through the tax rate hands most of it to high-PLI households who save it: the gap between the two closures is the iMPC–PLI covariance of Proposition 1, compounded over the life of the fiscal rule. In het-beta every household instrument reaches the same average MPC, so the composition of the rebate nearly washes out, and the only closure that stands apart is the one that bypasses household budgets altogether, government purchases, which enter demand one for one and change the size of the government rather than the incidence of a rebate.

¹⁵Het-beta's responses are larger under every closure because its labor-income iMPCs are higher (Figure 6, panel b), so the income-spending feedback amplifies any impulse more. The relevant comparison is across closures within each model.

5 The Effects of Rising Permanent Labor Income Inequality

The previous section compared models holding the distribution of permanent labor income fixed. This section considers the impact of a change in the distribution of PLI. We re-solve the model with the PLI distribution observed in 1970, keeping every other parameter at its baseline value and letting the real rate clear the asset market, and compare the two economies shock by shock. Under homothetic preferences every Jacobian is invariant to the PLI distribution (Proposition 4 in Appendix A), so that the aggregate response stays the same. In the baseline, the shift is far from neutral, for two reasons that Proposition 3 separates: each type behaves differently in the new steady state, and the types are weighted differently in the aggregate.

This section contains three main results. First, we show that rising PLI inequality raises the aggregate annual MPC from 0.30 to 0.36, raises the hand-to-mouth share from 0.31 to 0.39, and lowers the EIS from 0.81 to 0.75; between two thirds and four fifths of each change comes from the general-equilibrium fall in the real rate rather than from the income shift itself (Section 5.1). Second, with high PLI inequality, the incidence of fiscal policy matters even more: the gap between financing instruments widens for every payout, and the tax cut repaid by transfer cuts becomes more contractionary, falling from -0.49 to -0.61 (Section 5.2). Third, monetary policy gains a little power in the aggregate but changes composition: the direct rate channel shrinks, the income channels expand, the valuation gains at the top grow, and the output response becomes about 20% more sensitive to the fiscal rule (Section 5.3).

5.1 Rising inequality and the aggregate consumption function

The experiment runs as follows. Take two economies with the exact same calibration and change only the distribution of PLI: the baseline carries the 2019 distribution, the counterfactual the 1970 one. In each economy, the real rate adjusts so that the asset market clears, which delivers two steady states. We then compare how fiscal and monetary policy transmit in these two economies. Since the PLI distribution affects only the household block, the whole difference in transmission is captured by the

change in the household sequence-space Jacobians, and we start with the aggregate consumption function of Section 4.

Write per-capita consumption of type i as $C_{i,t}$, so that $C_t = \sum_i \omega_i C_{i,t}$, and define type-level Jacobians accordingly, e.g. $\mathbf{M}_i^\chi$ for the response of $C_{i,t}$ to transfers. Since a change dZ_s raises type- i per-capita labor income by $\alpha_i dZ_s$, define the *normalized* labor-income Jacobian $\widetilde{\mathbf{M}}_i^Z$ as the response of type- i per-capita consumption to a unit change in type- i per-capita post-tax labor income. Aggregation takes the form

$$\mathbf{M}^Z = \sum_i \omega_i \alpha_i \widetilde{\mathbf{M}}_i^Z, \quad \mathbf{M}^\chi = \sum_i \omega_i \mathbf{M}_i^\chi, \quad \mathbf{M}^r = \sum_i \omega_i \mathbf{M}_i^r, \quad \mathbf{m}^{cap} = \sum_i \omega_i \mathbf{m}_i^{cap} : \quad (19)$$

only the labor-income matrix carries income weights $\omega_i \alpha_i$; all other matrices aggregate with population weights. The comparison across steady states is organized by the following decomposition.

Proposition 3 (Comparing economies). *Consider two economies that share population shares ω_i but differ in their permanent income distribution α . For any object x , let x' denote its value in the second economy and $\Delta x = x' - x$. Each aggregate Jacobian difference splits into a composition and a behavior term:*

$$\Delta \mathbf{M}^Z = \underbrace{\sum_i \omega_i \Delta \alpha_i \widetilde{\mathbf{M}}_i^{Z'}}_{\text{composition}} + \underbrace{\sum_i \omega_i \alpha_i \Delta \widetilde{\mathbf{M}}_i^Z}_{\text{behavior}}, \quad \Delta \mathbf{M}^\chi = \sum_i \omega_i \Delta \mathbf{M}_i^\chi. \quad (20)$$

Proof. Appendix A. □

In our application, the first economy carries the 1970 distribution and the second the 2019 one, so Δ measures the change produced by rising inequality. Equation (20) says that a change in the PLI distribution moves the consumption function through two channels. The behavior term reflects that each type solves a different problem in the new steady state: its wealth, its distance to the borrowing constraint, and hence its policy functions all change. The composition term reflects that when PLI inequality rises, a larger share of aggregate labor income is paid to high-PLI, low-MPC households. Only the labor-income Jacobian $\mathbf{M}^Z$ carries this second margin; the population-weighted matrices $\mathbf{M}^\chi$, $\mathbf{M}^r$ and $\mathbf{m}^{cap}$ move through behavior alone.

The decomposition also delivers the neutrality anticipated at the end of Section 4. Under homothetic preferences ($\gamma = 0$, or $\gamma > 0$ with $\Sigma = 1$ and $\zeta = 0$) the type problems scale: the normalized labor-income and transfer Jacobians are common across types and across economies, and the rate and capital-gains Jacobians are proportional to α_i . Every aggregate Jacobian then depends on the distribution only through $\sum_i \omega_i \alpha_i = 1$, which is fixed by normalization, so all Jacobians are invariant across the two economies. Proposition 4 in Appendix A states the result formally: every Jacobian, and hence every output response of Section 4, is invariant to the PLI distribution α . This extends Lemma 2 from the incidence wedge to the full transmission mechanism and covers het-beta: what breaks neutrality is the non-homotheticity of the wealth term, not the taste for wealth itself.¹⁶ In the baseline, neither term is zero. The rest of this subsection measures both terms; the following ones re-run the fiscal and monetary experiments in the two economies.

The behavior effect. Figure 7 showed that the annual MPC is a decreasing and convex function of permanent labor income, while the EIS is increasing and concave. By Jensen's inequality, a mean-preserving spread of the PLI distribution therefore raises the population-average MPC and lowers the average EIS: the households whose permanent income falls slide up the steep part of the MPC schedule, and their MPC rises by more than the MPC of the households moving right along the flat part falls. The same convexity steepens the hand-to-mouth gradient: low-PLI households become more likely to hit the borrowing constraint, while high-PLI households were already unconstrained.

The composition effect. In $\mathbf{M}^Z$, the spread also shifts income weight from high-MPC toward low-MPC types, which offsets part of the behavior effect; $\mathbf{M}^X$ has no such offset. The gap between the two Jacobians therefore widens: the present-value wedge between a transfer dollar and a labor-income dollar rises from 0.13 to 0.17. Table 3 shows the four-quarter counterpart: the aggregate MPC rises from 0.30 to 0.36 while its income-weighted analogue rises only from 0.15 to 0.17, so the gap widens from 0.15 to 0.19. In het-beta, both objects are frozen by Proposition 4, and the present-

¹⁶One caveat applies to the quantitative model: equity exposure there follows the sigmoid rule of Equation (4) rather than being proportional to wealth, so the proposition applies to het-beta up to that departure, which is quantitatively negligible: the het-beta responses below are invariant to the second decimal.

value wedge stays at 0.02. Rising inequality thus amplifies precisely the mechanism of Section 4: the further apart transfers and labor income drift, the more the composition of a policy package matters.

Table 3: Aggregate HtM, annual MPC and EIS

	HtM	MPC (annual)	MPC income-weighted (annual)	EIS
Low inequality	0.31	0.3	0.15	0.81
High inequality - PE	0.33	0.32	0.15	0.8
High inequality - GE	0.39	0.36	0.17	0.75

Most of the change is general-equilibrium. Why do the policy functions move so much? When PLI inequality rises, high-PLI households receive a larger share of aggregate labor income and, because wealth is a luxury good, save most of it. For the asset market to clear, the annual real rate falls by about 70 basis points, from 2.7% to 2%. Low-PLI households then face both a lower permanent income and a lower return on saving: their target wealth falls toward the borrowing constraint. Table 3 decomposes the change along these lines. The partial-equilibrium row re-solves household behavior with the 2019 income distribution at 1970 prices: it accounts for only 2 of the 8 percentage-point rise in the hand-to-mouth share (0.31 to 0.33), 2 of the 6 points of the MPC increase, and 1 of the 6 points of the EIS decline. The remainder arrives with the fall in r . A model in which the real rate did not respond to the savings of high-PLI households would miss between two thirds and four fifths of the change in every aggregate that matters for policy transmission.

5.2 Fiscal policy

We re-run the six deficit-financed packages of Section 4.2, a payout through government spending, transfers, or a tax cut, each repaid through future transfer cuts or future tax increases, in both economies. Figure 10 reports the cumulative multipliers. Rising PLI inequality widens the gap between financing instruments for every payout: from 0.58 to 0.68 for spending, from 0.64 to 0.73 for transfers, and from 0.73 to 0.86 for the tax cut. The movements are asymmetric in an instructive way. Packages repaid through tax increases strengthen (the transfer payout rises from 0.82 to 0.93), because the repayment falls on high-PLI households who absorb it out of saving. The spend-

ing and tax-cut packages repaid through transfer cuts weaken (0.66 to 0.57 and -0.49 to -0.61), because the repayment now hits households with even higher MPCs; the transfer payout repaid by transfer cuts, whose payout and repayment share the same incidence, instead edges up (0.18 to 0.20) through the pure timing effect.

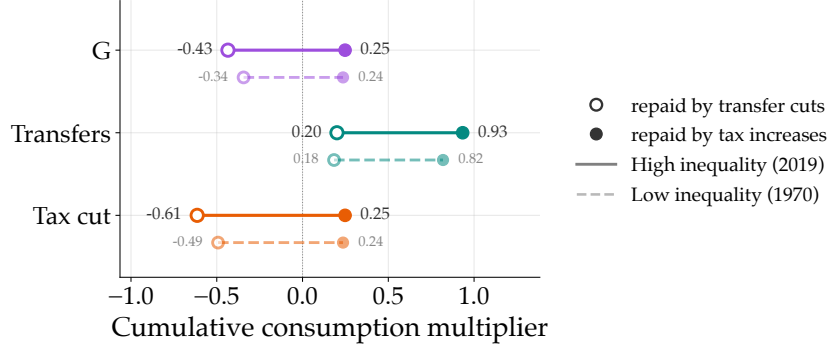


Figure 10: Cumulative multipliers by fiscal package: high against low PLI inequality

Note: Cumulative multipliers $(\sum_t Q^t dY_t) / (\sum_t Q^t d\epsilon_t)$, $Q = 1/(1+r)$, for deficit-financed packages combining a payout instrument (government spending, transfers, or a tax cut) with a financing instrument (future transfer cuts, hollow markers, or future tax increases, filled markers), in the high-inequality (solid) and low-inequality (dashed) calibrations. The figure reports consumption multipliers: they equal the output multipliers for the transfer and tax-cut payouts, and the output multiplier minus one for the spending payout.

The timing of the deficit matters more as well. For this exercise we step outside the fiscal rule (9) and follow Auclert et al. (2024): consider a transfer shock $d\chi_t = \rho_\chi^t d\chi_0$ with persistence $\rho_\chi = 0.9$, and let debt follow $dB_t = \rho_B (dB_{t-1} + d\chi_t)$, so that a higher ρ_B postpones the repayment further into the future. As shown by Auclert et al. (2024), the cumulative multiplier increases with ρ_B : near the borrowing constraint the Euler equation is discounted, in the sense of McKay et al. (2017) and Michailat & Saez (2021), so households spend today's payout faster than they cut spending for the repayment taken back later. Rising PLI inequality strengthens this force: with more households close to the constraint, iMPCs become more front-loaded and future repayments are discounted more heavily. Figure 11 shows the result: the cumulative multiplier rises with ρ_B in both economies, but the high-inequality economy gains more, and the gap itself grows with persistence, reaching 0.35 for tax-financed and about 0.5 for transfer-financed deficits as ρ_B approaches one. For transfer-financed deficits the gap only opens at high persistence: when the repayment arrives quickly, it hits the same high-MPC households as the payout, and the two nearly cancel regardless of the distribution.

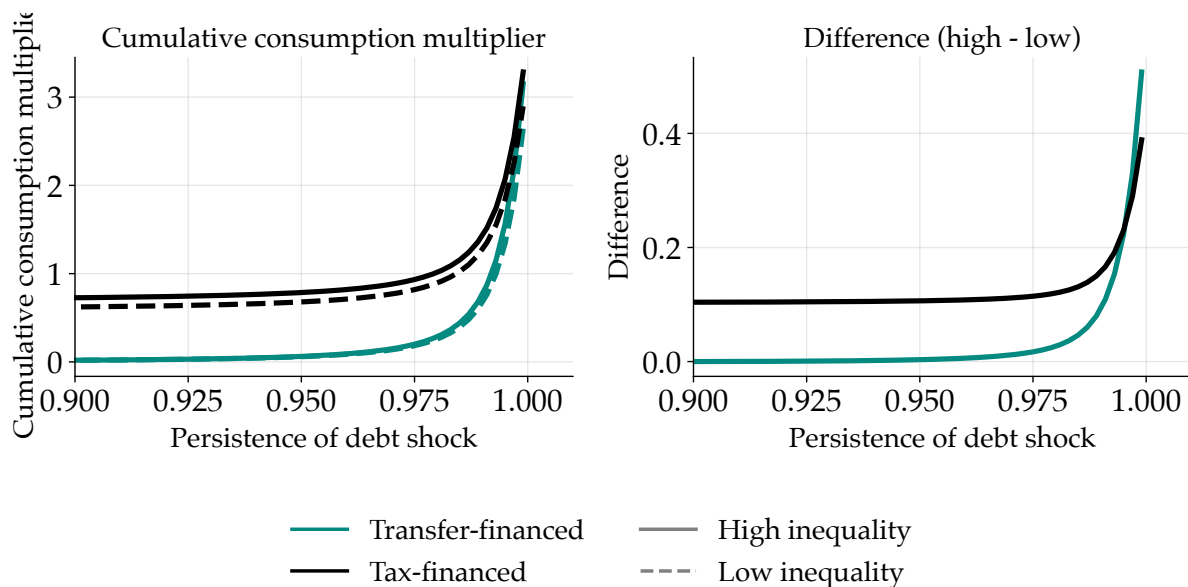


Figure 11: Cumulative multipliers and the persistence of the deficit

Note: The left panel shows the cumulative multiplier $(\sum_t Q^t dY_t) / (\sum_t Q^t d\chi_t)$, $Q = 1/(1+r)$, of a transfer shock as a function of the debt persistence ρ_B , when the deficit is repaid through transfers (green) or through taxes (black), in the high-inequality (solid) and low-inequality (dashed) calibrations. The right panel plots the difference between the high- and low-inequality curves.

5.3 Monetary policy

We turn to an expansionary monetary policy shock: an AR(1) fall in the quarterly real rate with an initial impulse of 25 basis points (about 100 basis points annualized) and persistence $\rho^r = 0.9$, with transfers adjusting so that the government's interest-inclusive obligation $(1 + r_{t+1})B_t$ stays at its steady-state value (debt itself moves as the rate falls). Following Equation (12), the consumption response splits into the direct effect of dr , the indirect effects operating through labor income dZ and through the government budget $d\chi$, and the capital-gains channel. Figure 12 reports this decomposition in the two economies, and the distribution of the income the shock generates on impact.

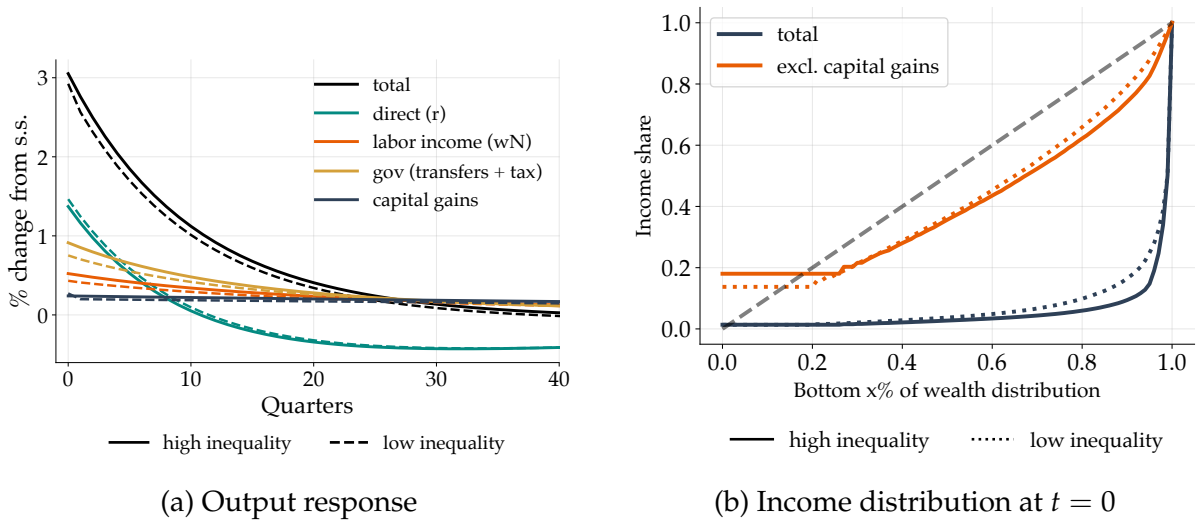


Figure 12: Output and income-distribution response to a monetary policy shock

Note: The left panel (a.) decomposes the output response to an expansionary monetary policy shock of 25 quarterly basis points with persistence 0.9 into the direct rate channel, the labor-income channel, the government channel (transfers and taxes), and capital gains. Full lines: high-inequality calibration; dashed lines: low-inequality calibration. The right panel (b.) shows the Lorenz curve of the income generated in the first period of the shock, dY_0 , across the wealth distribution, with (blue) and without (orange) capital gains.

A slightly stronger, differently composed response. The total output response is about 4% larger in the high-inequality economy on impact (2.44% against 2.34%). The composition shifts by much more than the total. The direct rate channel falls, the mechanical consequence of the lower aggregate EIS: with more households at the borrowing constraint, fewer reallocate consumption in response to intertemporal prices. The two income channels expand by about a fifth each: higher MPCs turn the labor income and the fiscal transfers generated by the easing into more spending. Rising inequality thus tilts monetary transmission away from intertemporal substitution and toward income-based feedbacks. This composition is consistent with the cross-sectional evidence: Cloyne et al. (2020) show that the consumption response to rate changes in the U.S. and the U.K. is driven by households with little liquid wealth, with the income effect dominating the direct cash-flow effect; Slacalek et al. (2020) reach the same conclusion for the euro area; and Almgren et al. (2022) find larger output responses to ECB shocks in countries with more hand-to-mouth households.

More regressive. The income the easing creates at $t = 0$ is also distributed more unequally. Figure 12b plots the Lorenz curve of dY_0 across the wealth distribution: capital gains dominate the date-0 income flow, they scale with wealth, and wealth is more concentrated in the high-inequality economy, so the same valuation channel delivers

more income to the top. Figure 13 expresses the consequence in dollars: on impact, the average household in the top 1% gains about 2.5 million USD in the high-inequality economy, against about 1.9 million in the low-inequality one. In the top 0.1% the two economies are close (6.8 against 7.2 million), because the 1970 economy holds a slightly larger top-0.1% wealth share (10% against 8%). Monetary easing was already regressive at the top of the wealth distribution in 1970; rising PLI inequality scales the gains of the top 1% up by about a third.

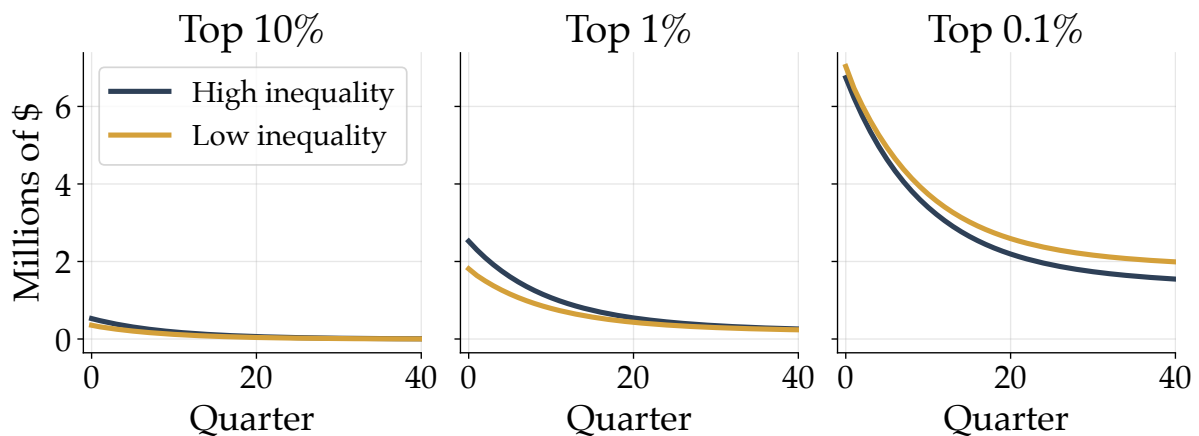


Figure 13: Wealth gains at the top following a monetary policy shock

Note: Increase in the average wealth of a household within the top 10%, top 1%, and top 0.1% of the wealth distribution following the monetary policy shock, in millions of USD, in the high- and low-inequality calibrations.

The fiscal rule matters more. A rate cut improves the government budget: it lowers the interest bill on public debt and raises tax receipts. We now replace the constant-obligation closure of Figures 12 and 13 with the fiscal rule of Equation (9), which decides what happens to this windfall along two margins: the instrument that returns it, transfers, the tax rate, or government spending, and the speed φ at which debt comes back to steady state, hence how soon the windfall reaches the economy. Figure 14 sweeps both at once: the left panel plots the annual consumption semi-elasticity of the same monetary shock for each instrument as a function of φ , the right panel the difference between the high- and low-inequality economies.

Start with the speed. Under every instrument the semi-elasticity rises with φ , from between 7.4 (spending) and 7.8 (transfers, taxes) when the government sits on the windfall (φ near zero) to between 8.7 (spending) and 10.4 (transfers) at $\varphi = 0.25$: over this range, the speed moves the output response by more than either the instrument

or the PLI distribution. The reason is the timing logic of Section 5.2 in reverse: a windfall handed back quickly lands on front-loaded iMPCs and is spent; a windfall used to pay down debt returns so far in the future that households discount it heavily. The instrument matters more the faster the windfall returns: transfers generate the largest response over the whole range beyond $\varphi \approx 0.01$, the tax rate comes second, since tax cuts accrue mostly to high-PLI households who save them, and spending is weakest, because it turns the windfall into demand one for one without any multiplier on household income.

Rising PLI inequality increases the sensitivity of the output response along both margins. At φ close to zero the difference between the two economies is slightly negative under all three rules, about -0.1 : with the windfall parked in the budget, monetary policy works mostly through the direct rate channel, which shrinks as inequality rises. The transfers difference turns positive at $\varphi \approx 0.07$, just above the baseline speed, and climbs to 0.21 at $\varphi = 0.25$, while the tax and spending differences stay slightly negative (-0.05 and -0.10). The extra power that inequality lends to monetary policy is therefore not intrinsic to the rate channel: it exists only when the windfall returns quickly, and through transfers, so that the fiscal flows land on households with higher MPCs. The gap between instruments widens as well: at $\varphi = 0.25$ the distance between the transfers and the tax rule grows from 1.15 to 1.41 , about 20% . A fiscal authority that saves the windfall, or rebates it through the tax rate, forgoes most of the extra power; rising inequality strengthens the case for fiscal-monetary coordination.

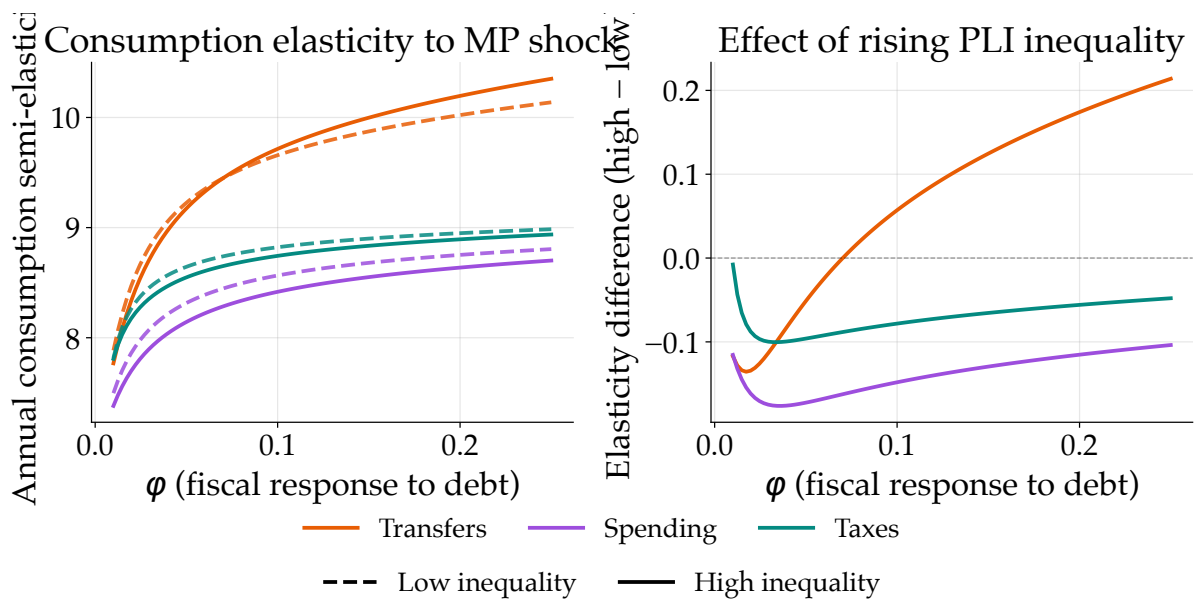


Figure 14: Consumption elasticity to a monetary policy shock, by fiscal rule

Note: The left panel shows the annual consumption semi-elasticity to the monetary policy shock as a function of the fiscal-rule speed ϕ , when the fiscal windfall is returned through transfers, the tax rate, or government spending, in the high-inequality (solid) and low-inequality (dashed) calibrations. The right panel plots the difference between the high- and low-inequality curves for each rule.

Conclusion

This paper asks whether the rise in permanent labor income inequality since 1970 has changed the transmission of monetary and fiscal policy. In standard heterogeneous-agent models, the answer is no: under homothetic preferences, the PLI distribution is neutral for the output response to any shock. The PSID suggests otherwise: hand-to-mouth status is concentrated among households with low permanent labor income, and this concentration has grown as inequality rose. A non-homothetic taste for wealth reproduces these facts and breaks the neutrality: high-permanent-income households save to accumulate wealth they value for its own sake, the equilibrium interest rate falls, and low-permanent-income households dissave toward the borrowing constraint.

Once the hand-to-mouth households are the low-permanent-income ones, the incidence of policy becomes a first-order determinant of its effect. Holding the deficit path fixed, switching the repayment instrument moves cumulative multipliers by more than 0.5, and a tax cut repaid by transfer cuts contracts output; the output effect of a rate cut varies by 98% with the instrument that rebates the government's interest sav-

ings. Rising PLI inequality strengthens these forces. Between the 1970 and the 2019 income distributions, the aggregate MPC rises from 0.30 to 0.36, mostly through the general-equilibrium fall in the real rate; fiscal packages financed by taxes strengthen while packages financed by transfers weaken; and monetary policy becomes about 4% stronger, more regressive at the top of the wealth distribution, and 20% more sensitive to the fiscal rule. The composition of a fiscal package, and the coordination between fiscal and monetary policy, matter more today than fifty years ago.

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A Derivations of the Sequence-Space IKC

This appendix derives the intertemporal Keynesian cross and proves the results of Sections 4 and 5. Throughout, all derivatives are evaluated at the steady state, $\mathbf{q}_t = (1 + r)^{-t}$, and we abstract from the bequest flow, whose response is proportional to the quarterly death rate ζ .

A.1 From the consumption function to the IKC

With $w = 1/\mu$ and $N_t = Y_t$, post-tax labor income satisfies

$$d\mathbf{Z} = \frac{1}{\mu} \left[(1 - \tau) d\mathbf{Y} - Y d\tau \right] = \frac{1}{\mu} (d\mathbf{Y} - d\mathbf{T}), \quad (21)$$

where $d\mathbf{T} = \tau d\mathbf{Y} + Y d\tau$ is the change in tax receipts. Since dividends are taxed at the same rate as labor, after-tax dividends are $(1 - \tau_t) \frac{\mu-1}{\mu} Y_t = \frac{\mu-1}{\mu} (Y_t - T_t)$: post-tax income $Y_t - T_t$ splits between wages and dividends in the constant shares $\frac{1}{\mu}$ and $\frac{\mu-1}{\mu}$. Differentiating the no-arbitrage condition (6) and the mutual fund balance sheet (11) yields the following lemma.

Lemma 1 (Capital gains). *To first order,*

$$dcap = \frac{\mu-1}{\mu} \mathbf{q}' (d\mathbf{Y} - d\mathbf{T}) - p \mathbf{q}' d\mathbf{r}, \quad (22)$$

with $dr_0 = 0$, since the date-0 return is predetermined.

Proof. Iterating the no-arbitrage condition (6) forward,

$$p_0 = \sum_{k \geq 1} \left[\prod_{j=1}^k (1 + r_j)^{-1} \right] (1 - \tau_k) d_k,$$

where r_j denotes the return credited at date j , so that the discount between 0 and k uses $r_1, \dots, r_k$. After-tax dividends satisfy $(1 - \tau_k) d_k = \frac{\mu-1}{\mu} (Y_k - T_k)$, so

$$d[(1 - \tau_k) d_k] = \frac{\mu-1}{\mu} (dY_k - dT_k),$$

and differentiating p_0 with respect to the after-tax dividend at $k \geq 1$ gives the discount

factor $(1+r)^{-k}$. For the interest rate, using $\partial[\prod_{j \leq k}(1+r_j)^{-1}]/\partial r_j = -(1+r)^{-k-1}$ for $j \leq k$,

$$\frac{\partial p_0}{\partial r_j} = -\sum_{k \geq j} (1+r)^{-k-1} (1-\tau) d = -(1-\tau) d \frac{(1+r)^{-j}}{r} = -p (1+r)^{-j}, \quad j \geq 1,$$

where the last equality uses the steady-state valuation $p = (1-\tau)d/r$. The mutual fund balance sheet (11) defines the aggregate capital gain as the unexpected excess value of equity,

$$dcap = \bar{S} dcap_0 = dp_0 + d[(1-\tau_0)d_0].$$

Adding the date-0 dividend term extends the sum to $k \geq 0$ and yields Equation (22), with $dr_0 = 0$ since the date-0 return was set at $t = -1$. $\square$

Substituting Equation (21) and Lemma 1 into Equation (12),

$$d\mathbf{C} = \frac{1}{\mu} \mathbf{M}^Z (d\mathbf{Y} - d\mathbf{T}) + \mathbf{m}^{cap} \left[\frac{\mu-1}{\mu} \mathbf{q}' (d\mathbf{Y} - d\mathbf{T}) - p \mathbf{q}' d\mathbf{r} \right] + \mathbf{M}^\chi d\chi + \mathbf{M}^r d\mathbf{r},$$

and collecting the $(d\mathbf{Y} - d\mathbf{T})$ and $d\mathbf{r}$ terms gives

$$d\mathbf{C} = \underbrace{\left[\mathbf{M}^r - p \mathbf{m}^{cap} \mathbf{q}' \right]}_{=\bar{\mathbf{M}}^r} d\mathbf{r} + \underbrace{\left[\frac{1}{\mu} \mathbf{M}^Z + \frac{\mu-1}{\mu} \mathbf{m}^{cap} \mathbf{q}' \right]}_{=\mathbf{M}^T} (d\mathbf{Y} - d\mathbf{T}) + \mathbf{M}^\chi d\chi, \quad (23)$$

the analogue of equation 9 in Auclert et al. (2025), with the combined matrices $\mathbf{M}^T$ and $\bar{\mathbf{M}}^r$ defined in Section 4.

For the government budget, differentiating Equation (8),

$$dB_t = (1+r) dB_{t-1} + B dr_t + dG_t + d\chi_t - dT_t,$$

which rearranges to Equation (16) in the main text. As in Auclert et al. (2025), fiscal policy sets a path for government debt and for all instruments but one, and the remaining instrument adjusts so that Equation (16) holds period by period.¹⁷ The debt

¹⁷The feedback rule (9) used in the quantitative model is one way of generating such a debt path; the propositions condition on the path itself.

term is present-value neutral by telescoping:

$$\begin{aligned}\mathbf{q}'[\mathbf{I} - (1+r)\mathbf{L}]d\mathbf{B} &= \sum_{t \geq 0} (1+r)^{-t} [dB_t - (1+r)dB_{t-1}] \\ &= \lim_{T \rightarrow \infty} (1+r)^{-T} dB_T - (1+r)dB_{-1} = 0,\end{aligned}$$

since $dB_{-1} = 0$ and debt returns to its steady state.

Combining Equation (23) with goods market clearing $d\mathbf{Y} = d\mathbf{C} + d\mathbf{G}$ gives the intertemporal Keynesian cross, Equation (15), with $d\mathbf{T}$ and one of $(d\chi, d\tau)$ tied down by Equation (16).

Invertibility. In the benchmark of Auclert et al. (2024), each household's intertemporal budget constraint implies that a windfall is entirely spent in present value, so discounted iMPCs sum to one, $\mathbf{q}'\mathbf{M}^T = \mathbf{q}'$, and $\mathbf{I} - \mathbf{M}^T$ is singular; the equilibrium is then selected by the government budget constraint. This is a property of the budget set, not of preferences. In our model the identity fails by an accounting choice: our Jacobians hold the bequest flow received by newborns fixed, so the estates of dying households, which in equilibrium return to households as newborn endowments, do not re-enter $\mathbf{M}^T$; with the bequest feedback included, the unit-present-value identity would be restored. Quantitatively, the leakage is of order ξ per period and cumulates over the spending horizon: $\mathbf{q}'\mathbf{M}_{\cdot 0}^\chi = 0.85$ in the baseline, so $\mathbf{q}'\mathbf{M}^T < \mathbf{q}'$ and the multiplier operator of Proposition 2 is invertible; we verify this numerically.

A.2 Proof of Proposition 1 (incidence and iMPCs)

A change dX_s raises the date- s income of household (a, e, i) by $\iota_s dX_s$, and the household-level response $m_{t,s}$ of date- t consumption per unit of date- s income does not depend on the label of the income. By linearity of aggregation,

$$\frac{\partial \mathcal{C}_t}{\partial X_s} = \int \iota_s m_{t,s} d\Phi = \mathbb{E}[m_{t,s}] + \text{Cov}(\iota_s, m_{t,s}),$$

using $\mathbb{E}[\iota_s] = 1$; the first term is the uniform-incidence Jacobian $\mathbf{M}_{t,s}^\chi$, where the expectation is over the cross-section of households at date s along the steady state. For labor

income, $\iota_s = \alpha_i e_s$: splitting the expectation by type and using $\mathbb{E}_i[e_s] = 1$,

$$\mathbb{E}[m_{t,s} \alpha_i e_s] - \mathbb{E}[m_{t,s}] = \sum_i \omega_i \alpha_i (\bar{m}_{i,t,s} + \text{Cov}_i(m_{t,s}, e_s)) - \sum_i \omega_i \bar{m}_{i,t,s},$$

and collecting terms, using $\sum_i \omega_i (\alpha_i - 1)$ on the mean part, gives Equation (13). For capital gains, $\iota_0 = s(a_{ss}, A_{ss})/\bar{S}$ at date 0 gives Equation (14). $\square$

A.3 Homothetic irrelevance of permanent income

The wedge result stated in prose in Section 4.1:

Lemma 2 (Irrelevance of permanent income under homothetic preferences). *Assume $\gamma = 0$ so that preferences are homothetic. Then $\bar{m}_{i,t,s}$ and $\text{Cov}_i(m_{t,s}, e_s)$ are independent of i : the iMPC–PLI covariance in Equation (13) is zero, and the wedge $\mathbf{M}^Z - \mathbf{M}^X$ reduces to the iMPC–transitory covariance, which is invariant to the PLI distribution α .*

Proof. With $\gamma = 0$ the objective is logarithmic in consumption, so scaling consumption by α_i adds the constant $\log \alpha_i$ to period utility and leaves choices unchanged, and the steady-state type- i problem scales in α_i : dividing the budget constraint (3) by α_i , post-tax labor income is proportional to α_i , financial income $(1+r)a$ scales with a , the newborn bequest is proportional to α_i , steady-state transfers and capital gains are zero (so neither χ_{ss} nor the equity-exposure rule enters), and the borrowing limit $a \geq 0$ is invariant to scale. Hence $c_i(a, e) = \alpha_i c^*(a/\alpha_i, e)$ and $\Phi_i(a, e) = \Phi^*(a/\alpha_i, e)$, where starred objects belong to the problem of a type with $\alpha = 1$. The income kernel $m_{t,s}$ is computed from these steady-state policies, so the exposure rule, which matters only for the incidence of the unanticipated capital gain, does not affect it. The income kernel inherits the scaling: a unit income payout raises type- i cash on hand by one unit, and differentiating $c_i(a, e) = \alpha_i c^*(a/\alpha_i, e)$ with respect to cash on hand cancels the α_i factors, so $m_{t,s}(a, e, i) = m_{t,s}^*(a/\alpha_i, e)$. Averaging over the stationary distribution,

$$\bar{m}_{i,t,s} = \int m_{t,s}^*(a/\alpha_i, e) d\Phi^*(a/\alpha_i, e) = \bar{m}_{t,s}^*,$$

independent of i , and likewise $\text{Cov}_i(m_{t,s}, e_s) = \text{Cov}^*(m_{t,s}^*, e_s)$. The iMPC–PLI covariance in Equation (13) is then $\bar{m}_{t,s}^* \sum_i \omega_i (\alpha_i - 1) = 0$ by the normalization $\sum_i \omega_i \alpha_i = 1$, and the iMPC–transitory covariance equals $\text{Cov}^*(m_{t,s}^*, e_s)$, which does not depend on

α . $\square$

A.4 Proof of Proposition 2 (fiscal policy) and Equation (18)

We derive the general case with a rate path; Proposition 2 is $d\mathbf{r} = 0$ and Equation (18) is $d\epsilon^x = 0$. Measure each instrument in payout units, $d\epsilon^\chi = d\chi$, $d\epsilon^G = dG$, $d\epsilon^T = -Y d\tau$, spent at $\mathbf{M}^\chi$, $\mathbf{M}^G \equiv \mathbf{I}$, and $\mathbf{M}^T$ respectively. Substituting receipts $d\mathbf{T} = \tau d\mathbf{Y} + Y d\tau$ into the IKC (15),

$$d\mathbf{Y} = (1 - \tau)\mathbf{M}^T d\mathbf{Y} + \mathbf{M}^\chi d\epsilon^\chi + \mathbf{M}^T d\epsilon^T + \mathbf{M}^G d\epsilon^G + \overline{\mathbf{M}}^r d\mathbf{r} :$$

every instrument enters through its own Jacobian, and the endogenous receipts $\tau d\mathbf{Y}$ reduce post-tax income spent at $\mathbf{M}^T$. In the same units, the budget identity (16) states that total payouts equal endogenous receipts, plus the interest saved on public debt, plus new borrowing:

$$d\epsilon^\chi + d\epsilon^T + d\epsilon^G = \tau d\mathbf{Y} - B d\mathbf{r} + [\mathbf{I} - (1 + r)\mathbf{L}] d\mathbf{B}.$$

The financing instrument absorbs the residual,

$$d\epsilon^f = \tau d\mathbf{Y} - B d\mathbf{r} + [\mathbf{I} - (1 + r)\mathbf{L}] d\mathbf{B} - \sum_{x \neq f} d\epsilon^x.$$

Substituting into the first display and collecting terms, using $(1 - \tau)\mathbf{M}^T + \tau\mathbf{M}^f = \mathbf{M}^T + \tau(\mathbf{M}^f - \mathbf{M}^T)$,

$$\begin{aligned} [\mathbf{I} - \mathbf{M}^T - \tau(\mathbf{M}^f - \mathbf{M}^T)] d\mathbf{Y} &= [\overline{\mathbf{M}}^r - B \mathbf{M}^f] d\mathbf{r} + \mathbf{M}^f [\mathbf{I} - (1 + r)\mathbf{L}] d\mathbf{B} \\ &\quad + \sum_{x \neq f} (\mathbf{M}^x - \mathbf{M}^f) d\epsilon^x, \end{aligned}$$

which gives Equation (17) at $d\mathbf{r} = 0$ with a payout on a single instrument, several payouts adding their corrections by linearity, and Equation (18) for a rate path with no fiscal payouts; uniqueness follows from invertibility of the multiplier operator. $\square$

A.5 Proof of Proposition 3 (decomposition)

Apply Equation (19) to each aggregate Jacobian: for the labor-income matrix,

$$\Delta \mathbf{M}^Z = \sum_i \omega_i \Delta(\alpha_i \widetilde{\mathbf{M}}_i^Z) = \sum_i \omega_i \Delta \alpha_i \widetilde{\mathbf{M}}_i^{Z'} + \sum_i \omega_i \alpha_i \Delta \widetilde{\mathbf{M}}_i^Z,$$

which is exact since $\Delta \alpha_i \widetilde{\mathbf{M}}_i^{Z'} + \alpha_i \Delta \widetilde{\mathbf{M}}_i^Z = \alpha_i' \widetilde{\mathbf{M}}_i^{Z'} - \alpha_i \widetilde{\mathbf{M}}_i^Z$ and population shares ω_i are held fixed, while the population-weighted matrices contribute only behavior terms, e.g. $\Delta \mathbf{M}^\chi = \sum_i \omega_i \Delta \mathbf{M}_i^\chi$.¹⁸ $\square$

A.6 Neutrality (Proposition 4): statement and proof

The neutrality of aggregate consumption with respect to the permanent income distribution is due to Straub (2019); the proposition extends it to every sequence-space Jacobian of the household block, and hence to the full transmission of fiscal and monetary shocks.

Proposition 4 (Neutrality). *Suppose preferences are homothetic ($\gamma = 0$, or $\gamma > 0$ with $\Sigma = 1$ and $\zeta = 0$), and equity exposure is proportional to wealth. Then the type problems scale: $c_i(a, e) = \alpha_i c^*(a/\alpha_i, e)$ and $\Phi_i(a, e) = \Phi^*(a/\alpha_i, e)$. Consequently $\mathbf{M}^Z$, $\mathbf{M}^\chi$, $\mathbf{M}^r$ and $\mathbf{m}^{cap}$, and hence $\mathbf{M}^T$, $\overline{\mathbf{M}}^r$ and the output responses in Proposition 2 and Equation (18), are invariant to the permanent income distribution α .*

Proof. Consider the type- i household problem and guess $c_i(a, e) = \alpha_i c^*(a/\alpha_i, e)$, $a_i'(a, e) = \alpha_i a^{*'}(a/\alpha_i, e)$, where starred objects solve the problem of a type with $\alpha = 1$. Divide the budget constraint (3) by α_i : post-tax labor income is proportional to α_i , financial income $(1+r)a$ scales with a , capital gains scale with a when exposure is proportional to wealth, the bequest received by a newborn is proportional to α_i , transfers are $\chi_{ss} = 0$, and the borrowing limit $a \geq 0$ is invariant to scale. The rescaled budget set is therefore that of the $\alpha = 1$ household. For preferences: with $\gamma = 0$, $v \equiv 0$ and the objective is logarithmic in consumption, so scaling consumption by α_i adds the constant $\log \alpha_i$ to period utility without affecting choices.¹⁹ The guess then verifies the

¹⁸The scalars p and B enter the operators through $\mathbf{M}^T$, $\overline{\mathbf{M}}^r$ and the fiscal channel; they differ slightly across steady states because aggregate wealth adjusts, and we fold these differences into the behavior effect when measuring the decomposition.

¹⁹The same argument goes through with $\gamma > 0$ if the wealth term is itself logarithmic, $\Sigma = 1$ and $\zeta = 0$, since then $u(\alpha c) = u(c) + \log \alpha$ and $v(\alpha a) = v(a) + \gamma \log \alpha$.

Bellman equation (2), and since the discount-factor process, the idiosyncratic process, and the rebirth rule are independent of type (newborn wealth scales with α_i through the bequest), the stationary distribution satisfies $\Phi_i(a, e) = \Phi^*(a/\alpha_i, e)$.

The scaling extends to perturbations. A change dZ_s perturbs type- i income by $\alpha_i e_s dZ_s$, a scaled version of the perturbation faced by the $\alpha = 1$ household; the induced consumption response is α_i times the normalized response per unit of the normalized perturbation, so $\widetilde{\mathbf{M}}_i^Z = \widetilde{\mathbf{M}}^{Z*}$ for all i . A transfer $d\chi_s$ is a scale-free income payout; by linearity of the first-order response, type i responds as α_i households of type $\alpha = 1$ each receiving $d\chi_s/\alpha_i$, so $\mathbf{M}_i^\chi = \mathbf{M}^{\chi*}$. A rate change perturbs income by $a dr_s$, which scales with α_i under the stationary distribution, and changes intertemporal prices identically for all types, so $\mathbf{M}_i^r = \alpha_i \mathbf{M}^{r*}$; similarly $\mathbf{m}_i^{cap} = \alpha_i \mathbf{m}^{cap*}$ under wealth-proportional exposure. Aggregating with Equation (19) and $\sum_i \omega_i \alpha_i = 1$,

$$\mathbf{M}^Z = \widetilde{\mathbf{M}}^{Z*}, \quad \mathbf{M}^\chi = \mathbf{M}^{\chi*}, \quad \mathbf{M}^r = \mathbf{M}^{r*}, \quad \mathbf{m}^{cap} = \mathbf{m}^{cap*},$$

none of which depend on α ; the combined matrices $\mathbf{M}^T$ and $\overline{\mathbf{M}}^r$ inherit the invariance. Since the scalars (τ, μ, B, p) are primitives or pinned down independently of α , the operators and impulses in Proposition 2 and Equation (18) are invariant, and so are the output responses. $\square$

B Robustness of the HtM–PLI Gradient

Figure 15 shows that the HtM–PLI gradient documented in Section 1 is not an artifact of a particular PLI construction. We recompute the gradient under three alternatives to the baseline Straub (2019) averaged residuals: (i) the original Aguiar et al. (2025) residuals (age cubic + year only), (ii) a household fixed-effects estimator with the Straub (2019) first-stage controls, and (iii) an empirical Bayes shrinkage estimator (Appendix D) that uses the estimated income process parameters ($\hat{\rho} = 0.91, \hat{\sigma}_\alpha^2 = 0.28$) to optimally weight each household’s average residual against the prior. All three yield a steep, near-monotonically declining gradient from decile 1 to decile 10, with remarkably similar shapes.

The gradient is also robust to the threshold defining hand-to-mouth status. Figure 16

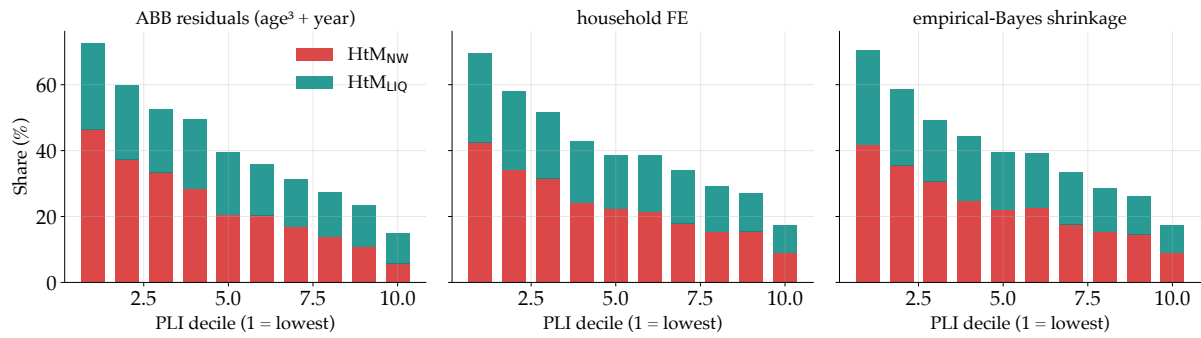


Figure 15: HtM shares by PLI decile: comparison across PLI measures

Notes: Left: Aguiar et al. (2025) residuals (age cubic + year FE). Center: household fixed-effects estimator with Straub (2019) controls. Right: empirical Bayes shrinkage estimator using MDE income process parameters. All panels use PSID 1999–2019, weighted.

recomputes HtM_{NW} shares by PLI quintile when the net-worth cutoff is halved to one month of earnings or raised to three months. Moving the threshold shifts the level of the hand-to-mouth share in every quintile, by more at the bottom than at the top, but leaves the relative gradient intact: the bottom-quintile share is about three times the top-quintile share at all three cutoffs (32 vs. 10 percent at one month, 38 vs. 12 at two, 42 vs. 14 at three).

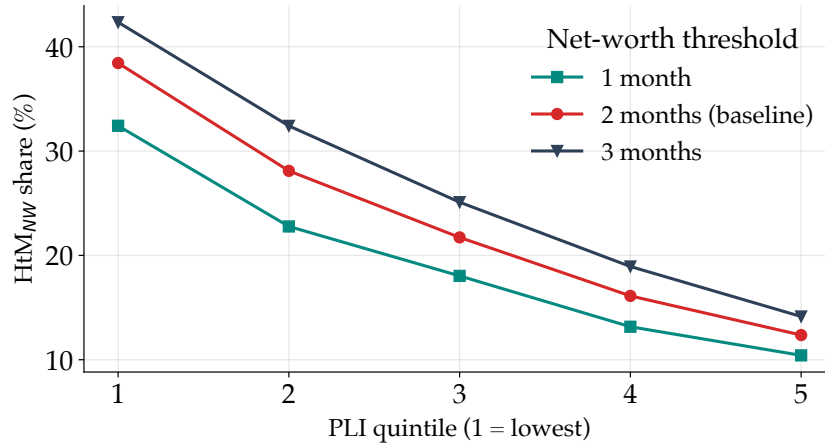


Figure 16: HtM_{NW} shares by PLI quintile: alternative net-worth thresholds

Notes: Weighted HtM_{NW} shares by PLI quintile when the hand-to-mouth cutoff is net worth below one, two (baseline), or three months of household earnings. Shares are computed among households with non-missing net worth and earnings. PSID 1999–2019.

C Additional Empirical Results

C.1 Summary Statistics

Table 4 reports weighted summary statistics by PLI quintile for the analysis sample of Section 1. Differences in permanent income map into large differences in balance sheets. Mean labor earnings rise from \$24,000 in the bottom quintile to \$156,000 in the top, median net worth from \$10,000 to \$268,000, and median liquid assets from essentially zero to \$22,000: the median household in the bottom two quintiles holds at most \$100 in liquid assets. Demographics, by contrast, are flat across quintiles. Mean age varies between 44.3 and 45.3 years and mean family size between 2.3 and 2.5, so the hand-to-mouth gradient of Section 1 does not reflect differences in age or household composition across permanent income groups.

Table 4: Summary statistics by PLI quintile (PSID, 1999–2019)

	Q1	Q2	Q3	Q4	Q5	All
Labor earnings (mean)	24.3	44.8	63.5	84.9	156.1	74.8
Nondurable consumption (mean)	23.0	30.2	35.9	43.4	58.8	38.3
Net worth (median)	10.0	36.4	74.3	120.9	267.6	72.3
Liquid assets (median)	0.00	0.10	0.92	4.96	22.31	0.95
Age (mean)	45.3	45.0	44.3	44.8	45.1	44.9
Family size (mean)	2.3	2.4	2.5	2.5	2.4	2.4
HtM _{NW} share (%)	38.3	27.8	22.1	15.9	12.4	23.3
HtM _{LIQ} share (%)	25.7	19.2	17.2	14.5	9.9	17.3
Observations	6,564	7,016	7,160	6,648	6,110	33,498
Households	1,262	1,177	1,154	1,027	945	5,565

Notes: Weighted by PSID family weights. Dollar amounts in thousands of 2012 dollars (PCE deflator). Nondurable consumption follows the 21-category aggregate of Aguiar et al. (2025). PLI quintiles based on the averaged residuals of Straub (2019). HtM shares are computed among households with non-missing hand-to-mouth status.

C.2 Rising Dispersion of Residual Earnings

Figure 17 plots the cross-sectional dispersion of residualized log household earnings by wave. The weighted standard deviation rises from 0.82 to 0.89 between the first and last waves, and the weighted P90–P10 gap from 1.96 to 2.26 log points. Most of the increase occurs between 2000 and 2008, with a plateau afterwards. Over its shorter window, the PSID thus displays the same widening of the labor income distribution as the DINA data used to calibrate the low- and high-inequality economies in Sec-

tion 3; because the PSID undersamples top earners, the increase at the very top of the distribution is likely understated here.

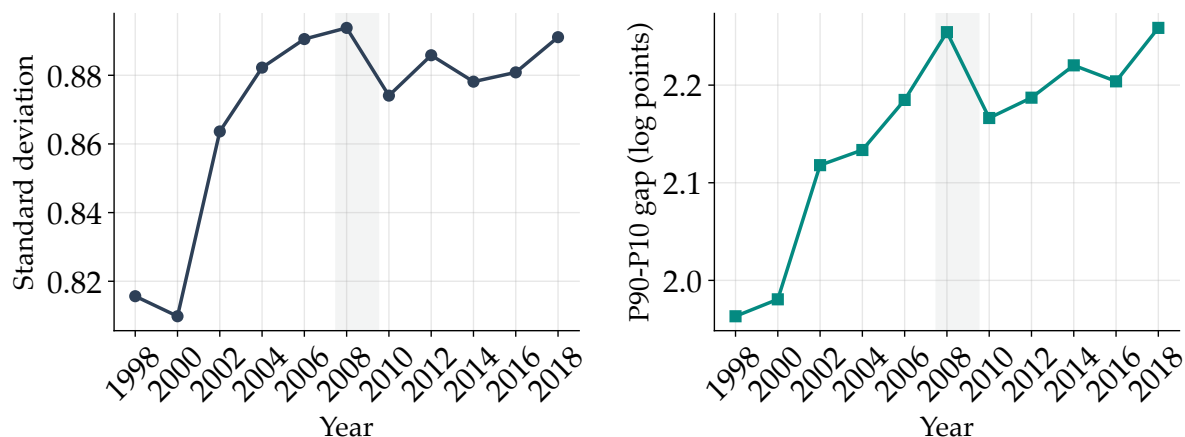


Figure 17: Dispersion of residualized log earnings over time (PSID, 1999–2019)

Notes: Weighted standard deviation (left) and weighted P90–P10 gap (right) of residualized log household labor earnings, by wave. Residuals from the Straub (2019) first stage (age cubic, family size, dual-earner indicator, year fixed effects). The vertical grey bar marks the Great Financial Crisis (2008–2009).

C.3 The Gradient Steepens with Inequality

Table 5 relates the steepening of Section 1 to inequality directly. We regress net-worth hand-to-mouth status on continuous PLI, its interaction with the wave-level dispersion of residual log earnings (demeaned), and year fixed effects. The interaction is negative and strongly significant under both dispersion measures: the HtM–PLI gradient is steeper in waves in which residual-earnings dispersion is higher, consistent with Fact 2.

C.4 Hand-to-Mouth Status, Permanent Income, and Consumption Dynamics

The gradient documented in Section 1 matters for policy transmission only if hand-to-mouth status and permanent income predict consumption behavior. Following Aguiar et al. (2025), Table 6 regresses biennial nondurable consumption growth on lagged hand-to-mouth status and demographic controls; Table 7 repeats the exercise for consumption volatility, measured as the absolute residual of consumption growth after removing household fixed effects.

Table 5: The HtM–PLI gradient and the level of inequality

	(1)	(2)
PLI	-0.148 (0.008)	-0.148 (0.008)
PLI $\times$ Ineq (std)	-0.761 (0.172)	
PLI $\times$ Ineq (P90–P10)		-0.247 (0.054)
R ²	0.06	0.06
N	30,795	30,795

Notes: Linear probability regressions of HtM_{NW} status on continuous PLI, its interaction with the wave-level dispersion of residual log earnings (demeaned; standard deviation in column 1, P90–P10 gap in column 2), and year fixed effects. The PLI coefficient is the gradient at average inequality. Weighted by PSID family weights; standard errors clustered by household.

Two results stand out. First, hand-to-mouth households display excess consumption sensitivity: lagged HtM_{NW} status predicts higher subsequent consumption growth (0.020 with household fixed effects, column b of Table 6) and higher consumption volatility (0.014, column a of Table 7). Second, permanent income predicts consumption dynamics beyond hand-to-mouth status. Adding PLI attenuates the HtM_{NW} volatility coefficient by roughly one third, and membership in the bottom PLI quintile is the strongest single predictor of consumption volatility, with or without controlling for hand-to-mouth status (0.023 and 0.026 respectively, SE 0.004). Low-permanent-income households are thus not only more likely to be hand-to-mouth; conditional on their balance sheet, their consumption is also more exposed to shocks. This motivates a model in which MPC heterogeneity is organized along the permanent income dimension rather than by wealth alone.

D Estimation of the Income Process

This appendix details the two procedures we use to estimate the income process in the PSID. A minimum-distance estimation recovers the population parameters of the process, the dispersion of permanent income and the persistence and size of income shocks, from the autocovariance structure of residual log earnings. An empirical Bayes shrinkage estimator then combines these parameters with each household’s own earnings history to produce household-level estimates of permanent labor income, used in

Table 6: Consumption growth: hand-to-mouth status vs. permanent income

	ABB OLS (a)	ABB FE (b)	+ PLI (c)	PLI only (d)
HtM _{NW} (lagged)	0.006 (0.004)	0.020 (0.007)	0.010 (0.004)	
HtM _{LIQ} (lagged)	-0.008 (0.003)	0.002 (0.005)	-0.004 (0.004)	
PLI			0.009 (0.002)	0.009 (0.002)
PLI ²			0.001 (0.002)	0.001 (0.001)
R ²	0.07	0.19	0.07	0.07
N	24,202	24,202	24,202	26,366
Fixed Effects	No	Yes	No	No

Notes: Dependent variable: biennial log growth of nondurable consumption (annualized). Columns: (a) Aguiar et al. (2025) OLS specification; (b) with household fixed effects; (c) adding continuous PLI and its square; (d) PLI only. Column (d) does not require lagged hand-to-mouth status and is therefore estimated on a larger sample than columns (a)–(c). All specifications include an age quadratic, family size and marital status dummies, a dual-earner indicator, and year fixed effects. Weighted by PSID family weights; standard errors clustered by household.

Table 7: Consumption volatility: hand-to-mouth status vs. permanent income

	ABB (a)	+ PLI (b)	PLI only (c)
HtM _{NW} (lagged)	0.014 (0.003)	0.010 (0.003)	
HtM _{LIQ} (lagged)	0.005 (0.003)	0.001 (0.003)	
PLI Q1		0.023 (0.004)	0.026 (0.004)
PLI Q2		0.005 (0.004)	0.007 (0.004)
PLI Q3		0.004 (0.003)	0.005 (0.003)
PLI Q4		0.001 (0.004)	0.002 (0.004)
R ²	0.02	0.02	0.02
N	24,202	24,202	24,202

Notes: Dependent variable: absolute residual of biennial nondurable consumption growth from a first-stage regression of consumption growth on the lagged hand-to-mouth indicators, the controls of Table 6, and household fixed effects (common across columns). Columns: (a) lagged hand-to-mouth indicators; (b) adding PLI quintile dummies (Q5 omitted); (c) PLI quintile dummies only. Weighted by PSID family weights; standard errors clustered by household.

the robustness checks of Appendix B. The two procedures answer different questions: the first asks how much permanent inequality there is, the second asks where each household sits in it.

D.1 Statistical model

Let $\hat{y}_{it}$ denote residual log labor earnings of household i in income year t : the residual from the first-stage regression of Section 1, which removes an age cubic, family-size dummies, a dual-earner indicator, and year fixed effects. Following Straub (2019), we assume the residual is the sum of four unobserved components,

$$\hat{y}_{it} = \alpha_i + \eta_{it} + \psi_{it} + \nu_{it}. \quad (24)$$

The permanent component α_i , with variance σ_α^2 , is drawn once per household and never changes; it is the empirical counterpart of permanent labor income in the main text. The persistent component follows an AR(1) at annual frequency, $\eta_{it} = \rho \eta_{i,t-1} + \varepsilon_{it}$ with innovation variance σ_η^2 , initialized at $\eta_{i0} = 0$ at the start of the sample. The transitory shock ψ_{it} , with variance σ_ψ^2 , is independent across households and years. Measurement error ν_{it} enters the moments exactly like ψ_{it} and is therefore not separately identified; we fix its variance at $\sigma_\nu^2 = 0.02$, following Heathcote et al. (2010). The free parameter vector is $\theta = (\sigma_\alpha^2, \rho, \sigma_\eta^2, \sigma_\psi^2)$.

D.2 Minimum-distance estimation

Empirical moments. The estimation sample covers the eleven biennial PSID waves with income years 1998–2018, restricted to households aged 30–55 with at least five earnings observations: 2,395 households. Index waves by $t = 1, \dots, 11$. For every wave pair (t, t') we compute the empirical autocovariance

$$G_{t,t'} = \frac{1}{N_{t,t'}} \sum_{i \in \mathcal{I}_{t,t'}} \hat{y}_{it} \hat{y}_{it'}, \quad (25)$$

where $\mathcal{I}_{t,t'}$ is the set of households observed in both waves and $N_{t,t'}$ its cardinality; since the residuals have mean zero by construction, the average cross-product estimates the covariance. We require $N_{t,t'} \geq 30$, which in our sample leaves all $11 \times$

12/2 = 66 distinct moments available.

Theoretical moments. The model implies, for wave t and lag $s \geq 0$ measured in waves,

$$\text{Cov}(\hat{y}_{it}, \hat{y}_{i,t+s}) = \sigma_\alpha^2 + \text{Var}(\eta_{it}) \rho^{2s} + \mathbf{1}\{s = 0\} (\sigma_\psi^2 + \sigma_v^2), \quad \text{Var}(\eta_{it}) = \sigma_\eta^2 \sum_{j=0}^{2t-1} \rho^{2j}, \quad (26)$$

where the exponents $2s$ and $2t$ translate wave distances into years, since waves are two years apart while the AR(1) runs at annual frequency. Each parameter leaves a distinct fingerprint in (26). The permanent variance σ_α^2 is the level at which autocovariances flatten out at long lags, where the persistent and transitory components have died away. The persistence ρ governs how fast autocovariances decay toward that floor as the lag grows. The innovation variance σ_η^2 governs how fast the within-cohort variance of earnings rises across waves as shocks accumulate from the $\eta_{i0} = 0$ initial condition. The sum $\sigma_\psi^2 + \sigma_v^2$ is the excess of the variance ($s = 0$) over adjacent covariances, and fixing σ_v^2 pins down σ_ψ^2 .

Estimator. The estimate $\hat{\theta}$ minimizes the equally weighted sum of squared deviations between the 66 empirical and theoretical moments,

$$\hat{\theta} = \arg \min_{\theta} \sum_{t \leq t'} (G_{t,t'} - g_{t,t'}(\theta))^2, \quad (27)$$

with $g_{t,t'}(\theta)$ given by (26). We weight moments equally rather than by an estimated optimal weighting matrix, which is known to produce badly biased estimates of covariance structures in samples of this size (Altonji & Segal 1996). With 66 moments and four parameters the model is heavily overidentified; the minimized objective is 0.017.

Estimates. Table 8 reports the estimates. Estimated persistence is nearly identical to Straub (2019), and the permanent variance is roughly twice as large, consistent with our later sample and different first-stage controls. The transitory variance is also about twice as large, but for a more mechanical reason, tied to the biennial spacing of our data, that we take up next. The estimates imply an unconditional standard deviation

Table 8: Minimum-distance estimates of the income process (annual frequency)

		This paper	Straub (2019)
Persistence	ρ	0.907	0.904
Variance, permanent component	σ_α^2	0.280	0.131
Variance, persistent innovation	σ_η^2	0.027	0.026
Variance, transitory shock	σ_ψ^2	0.098	0.052
Variance, measurement error	σ_v^2	0.020 (fixed)	—

Notes: Equally weighted minimum distance on the 66 autocovariances of residual log household labor earnings, PSID income years 1998–2018, households aged 30–55 with at least five observations (2,395 households). The last column reproduces the corresponding estimates in Straub (2019).

of the persistent component of $\sqrt{\sigma_\eta^2 / (1 - \rho^2)} = 0.39$, and a cross-sectional variance of residual earnings of $\sigma_\alpha^2 + \sigma_\eta^2 / (1 - \rho^2) + \sigma_\psi^2 + \sigma_v^2 = 0.55$. The permanent component thus accounts for 51 percent of the residual cross-sectional variance of log earnings: half of measured labor income inequality at a point in time reflects differences that never mean-revert.

Transitory variance and the biennial spacing. The transitory shock and the measurement error enter the autocovariances only on the diagonal, at lag $s = 0$, pooled together, so fixing σ_v^2 identifies σ_ψ^2 from the excess of the same-wave variance over the nearest observed covariance. In our panel that nearest covariance is two years away. A shock that mean-reverts within two years is then observationally equivalent to a purely transitory one, since both have decayed by the following wave, and the single AR(1) whose high persistence matches the long-lag decay cannot also represent such a fast-reverting component. Its variance is absorbed into the diagonal and attributed to σ_ψ^2 , which inflates the transitory estimate relative to what annual data would assign to it. This is why our estimate, $\sigma_\psi^2 = 0.098$, is about twice the 0.052 of Straub (2019), who estimates the process on the annual PSID waves of 1981–1997: the one-year autocovariance that separates a fast-reverting persistent shock from a transitory one is present in annual data but absent from ours. The model’s idiosyncratic process carries only the persistent component, which is well identified from the off-diagonal decay: e_t follows the AR(1) of Section 3, with persistence 0.91 and unconditional standard deviation 0.39. The transitory component and the measurement error are not carried into the model.

D.3 Empirical Bayes shrinkage estimator

The natural estimator of household i 's permanent income is its average residual $\bar{y}_i = T_i^{-1} \sum_t \hat{y}_{it}$ over its T_i observed waves, our baseline PLI measure in the main text. It is unbiased for α_i but noisy: each wave adds noise $\eta + \psi + \nu$ with variance $\sigma_u^2 = \sigma_\eta^2 / (1 - \rho^2) + \sigma_\psi^2 + \sigma_\nu^2 = 0.27$, so for a household observed few times the average is dominated by luck rather than by α_i .

The shrinkage estimator treats the minimum-distance estimates as a prior. Across households, α_i has mean zero and variance σ_α^2 ; household i 's data contribute the signal $\bar{y}_i$ with noise variance σ_u^2 / T_i . The posterior mean weighs the two by their precisions:

$$\hat{\alpha}_i = \lambda_i \bar{y}_i, \quad \lambda_i = \frac{\sigma_\alpha^2}{\sigma_\alpha^2 + \sigma_u^2 / T_i}. \quad (28)$$

Households with long panels keep most of their average (at the median $T_i = 6$, $\lambda_i = 0.86$), while households observed once or twice are pulled toward the population mean; the average shrinkage factor is 0.83. The logic is regression to the mean: a household observed twice with a very high average residual is more likely a middling household with two lucky draws than a genuinely extreme earner, so its raw average overstates its permanent income. Shrinking accepts a small bias toward zero in exchange for a large variance reduction, and minimizes the mean squared error of each household-level estimate.

Two properties of the estimator matter for how we use it. First, posterior means are individually optimal but collectively compressed: their cross-sectional variance is $\lambda_i \sigma_\alpha^2$ on average, below σ_α^2 . The shrinkage measure is therefore suited to ranking households and to regression use, where it reduces attenuation from measurement error, but not to estimating the dispersion of permanent income itself, for which we use σ_α^2 directly. Second, the noise formula in (28) treats the η draws as independent across waves. They are not: with biennial persistence $\rho^2 = 0.82$, a household's persistent component is highly correlated across waves, so averaging reduces its contribution more slowly than $1/T_i$. The true noise variance of $\bar{y}_i$ is therefore somewhat larger than σ_u^2 / T_i , and (28) shrinks less than the optimum; the estimator stays conservative, in the sense of remaining close to the unshrunk average.

E Analytical Approximation of Pareto Tails in the Model

Consider a household of permanent type i solving

$$V_i(a_{t-1}, e_t) = \max_{c_t, a_t \geq 0} \log(c_t) + \gamma \frac{(a_t + \zeta)^{1-\Sigma}}{1-\Sigma} + \beta(1-\xi) \mathbb{E}_t [V_i(a_t, e_{t+1})],$$

subject to

$$c_t + a_t = (1 - \tau_t) w N_t \alpha_i e_t + (1 + r_t) a_{t-1} + \chi_t,$$

and $a_t \geq 0$. Since the derivation focuses on the steady state, we assume $r_t = r$ and abstract from transfers ($\chi_t = 0$) and from the bequest endowment received at rebirth. The proof trivially extends when these are positive, as long as they are not correlated with individual wealth.

The Euler equation is

$$c_t^{-1} = (1 - \xi) \beta (1 + r) \mathbb{E}_t [c_{t+1}^{-1}] + \gamma (a_t + \zeta)^{-\Sigma}.$$

For large wealth $a_{t-1} \rightarrow \infty$, labor income becomes negligible and idiosyncratic shocks vanish from the limiting policy function. The problem simplifies to

$$V(a_{t-1}) = \log(c_t) + \gamma \frac{(a_t + \zeta)^{1-\Sigma}}{1-\Sigma} + \beta(1-\xi) V(a_t),$$

subject to

$$c_t + a_t = (1 + r) a_{t-1}.$$

The Euler equation thus becomes

$$c_t^{-1} = (1 - \xi) \beta (1 + r) c_{t+1}^{-1} + \gamma (a_t + \zeta)^{-\Sigma}.$$

We guess a power-law consumption function of the form

$$c_t = B a_{t-1}^{\Sigma}.$$

Then the asset accumulation equation becomes

$$a_t = (1 + r)a_{t-1} - B a_{t-1}^\Sigma.$$

As $a_{t-1} \rightarrow \infty$,

$$a_t \sim (1 + r)a_{t-1}.$$

Thus, asymptotically,

$$c_{t+1} = B a_t^\Sigma \sim B(1 + r)^\Sigma a_{t-1}^\Sigma.$$

Plugging into the Euler equation, with the taste term evaluated at $a_t \sim (1 + r)a_{t-1}$ and ζ negligible as $a_{t-1} \rightarrow \infty$:

$$B^{-1} a_{t-1}^{-\Sigma} = (1 - \zeta)\beta(1 + r) B^{-1} (1 + r)^{-\Sigma} a_{t-1}^{-\Sigma} + \gamma(1 + r)^{-\Sigma} a_{t-1}^{-\Sigma}.$$

Dividing both sides by $a_{t-1}^{-\Sigma}$ and rearranging gives:

$$B^{-1} = (1 - \zeta)\beta(1 + r)^{1-\Sigma} B^{-1} + \gamma(1 + r)^{-\Sigma}.$$

Since this equation is linear in B^{-1} , solving for B gives the closed form

$$B = \frac{1 - (1 - \zeta)\beta(1 + r)^{1-\Sigma}}{\gamma(1 + r)^{-\Sigma}}.$$

The law of motion for wealth at very high levels is

$$a_t \approx (1 + r)a_{t-1}.$$

With death probability ζ , the transition is:

1. with probability $(1 - \zeta)$: wealth grows by $(1 + r)$
2. with probability ζ : wealth becomes 0 (death and restart at low wealth)

Thus, the multiplicative growth factor is

$$G_t = \begin{cases} 1 + r, & \text{with prob. } (1 - \zeta), \\ 0, & \text{with prob. } \zeta. \end{cases}$$

Let $s = 0$ denote "alive" and $s = 1$ denote "dead". The conditional moment generating matrix is:

$$M(k) = \begin{pmatrix} (1+r)^k & 0 \\ (1+r)^k & 0 \end{pmatrix}.$$

The survival transition matrix is

$$P = \begin{pmatrix} 1 - \xi & \xi \\ 1 - \xi & \xi \end{pmatrix}.$$

The Beare–Toda condition for a Pareto tail is

$$\rho(P \circ M(k)) = 1,$$

where $\circ$ is the Hadamard product.

Thus

$$P \circ M(k) = \begin{pmatrix} (1 - \xi)(1+r)^k & 0 \\ (1 - \xi)(1+r)^k & 0 \end{pmatrix},$$

whose spectral radius is

$$\rho = (1 - \xi)(1+r)^k.$$

Setting $\rho = 1$ yields the Pareto exponent ζ^a :

$$(1 - \xi)(1+r)^{\zeta^a} = 1,$$

so

$$\boxed{\zeta^a = \frac{-\log(1 - \xi)}{\log(1+r)}}.$$

From the asymptotic consumption rule,

$$c_t = Ba_{t-1}^\Sigma,$$

we invert:

$$a_{t-1} = B^{-1/\Sigma} c_t^{1/\Sigma}.$$

Since a has a Pareto tail with parameter ζ^a , i.e.

$$P(a > x) \sim Cx^{-\zeta^a},$$

we obtain

$$P(c > y) = P\left(a > B^{-1/\Sigma}y^{1/\Sigma}\right) \sim C\left(B^{-1/\Sigma}y^{1/\Sigma}\right)^{-\zeta^a},$$

so

$$P(c > y) \sim C'y^{-\zeta^a/\Sigma}.$$

Thus consumption has a Pareto tail with exponent

$$\boxed{\zeta^c = \frac{\zeta^a}{\Sigma}}.$$

F Calibration of the Portfolio Shares

Wealthy households hold a larger fraction of their assets in equity and business wealth, whose valuation increases when the risk-free interest rate declines. We follow Auclert & Rognlie (2020) and capture this feature of the data with the exogenous portfolio rule of Equation (4), fitted to the equity and business wealth shares observed in the 2019 Survey of Consumer Finances. The fit yields $\psi_0 = 3.62$, $\psi_1 = 0.48$, and $\psi_2 = 6.91$. Figure 18 compares the fitted function to the data. In our main calibration, we rely on the narrow definition of capital, which counts direct equity holdings and private business wealth as capital and excludes housing.

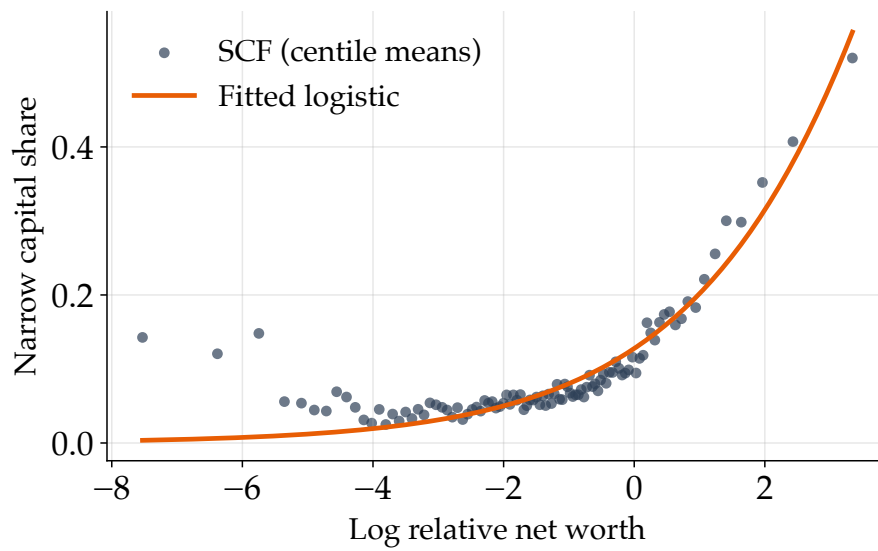


Figure 18: Portfolio Shares in the SCF 2019

Note: The dots show the share of equity and business wealth in total assets by wealth group in the 2019 Survey of Consumer Finances. The line shows the calibrated sigmoid portfolio-share function of Equation (4).